ELLIPTIC OBJECTS AND SUPERSYMMETRIC FIELD THEORIES

DAVID ARETZ AND DANIEL BERWICK-EVANS

Contents

1. Supersymmetric quantum mechanics and K-theory 2
2. Quantum field theories as representations of bordism categories 11
3. Supersymmetric field theories and topological modular forms 20
4. Towards equivariant elliptic objects and geometric cocycles for TMF 27
References 32

The primary goal of these lectures is to explain and give evidence for the following conjecture.

Conjecture A (Segal [Seg88], Stolz–Teichner [ST04, ST11])**.** The collection of 2-dimensional supersymmetric field theories determines geometric cocycles for (twisted equivariant) elliptic cohomology.

We provide a more specific phrasing in Conjecture E below. Beyond its intrinsic interest, work on this conjecture has a long track record of providing deep insights into mathematical quantum field theory. So in addition to unpacking the Segal–Stolz–Teichner program, a secondary goal of these lectures is to provide an onramp to quantum field theory for mathematicians.

Throughout, we indicate open problems. Sometimes, solutions are in the folklore but not documented anywhere (and for that reason, the folklore may be incomplete). These problems carry a single ($\star$). Problems that appear more difficult but still approachable with modern tools are labeled by ($\star\star$). Problems whose resolution will likely require genuinely new ideas are labeled by ($\star\star\star$). These gradings are of course subjective, based on incomplete data, and could well be wrong. Throughout, exercises are also provided for readers looking to deepen their understanding of the material.

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1. Supersymmetric quantum mechanics and K-theory

The mathematics of quantum field theory is still very much under development: we have many good ideas, but there is no single satisfactory framework. At least in part this is because quantum field theory is still a fluid concept in physics. Of course, many examples are well-understood in both communities, and we have many powerful tools for analyzing specific phenomena. But the larger thematic underpinnings still feel elusive. By contrast, the mathematical foundations of quantum mechanics are basically settled. This is not to say that we know everything about quantum mechanics: there are many interesting open questions! But these questions can (almost always) be couched in a rigorous mathematical framework.

Given this state of affairs, we will start our journey toward Conjecture A in the realm of quantum mechanics. This will allow us to build some intuition before taking the leap into quantum field theory. Another reason to start here is that supersymmetric quantum mechanics is a very useful prototype: we will see how it leads to a geometric cocycle model for (twisted, equivariant) K-theory [HST10].

1.1. Quantum mechanics.

Definition 1.1. Define the groupoid QM of *quantum mechanical systems* as having:

- Objects: Hilbert spaces $\mathcal{H}$ equipped with an (unbounded) self-adjoint operator $H\colon \mathcal{H} \to \mathcal{H}$.
- Morphisms: projective unitary operators $[u] \in \mathrm{PU}(\mathcal{H}, \mathcal{H}')$ such that $H' = uHu^{-1}$.

Various features of QM have standard names in physics. The operator H is the *Hamiltonian*. It generates unitary time evolution e^{itH}. After Wick rotation $t \rightsquigarrow it$, this "evolution" is through the contracting semigroup e^{-tH}. Nonzero vectors in $\mathcal{H}$ determine *states*, where two vectors represent the same state if they are scalar multiples of each other. Eigenvalues of H are *energy levels*, and *states of energy* λ comprise the λ-eigenspace of H. *Ground states* are the lowest-energy states. We will often deploy this terminology from physics below.

Example 1.2 (Quantum mechanics on a manifold)**.** Let M be a closed oriented Riemannian manifold. Define an object of QM by

$$(L^2(M), \Delta) \in \mathsf{QM}, \qquad \langle \psi, \psi' \rangle = \int_M \overline{\psi}\psi' d\mathrm{vol}, \ \psi, \psi' \in L^2(M)$$

i.e., the Hilbert space $\mathcal{H}$ is L^2-functions on M for the usual inner product, with Hamiltonian H given by the Laplacian Δ. An isometry $f\colon M \to M'$ of closed Riemannian manifolds induces the evident morphism in QM, giving a functor

$$\left\{ \begin{array}{c} \text{Closed Riemannian manifolds} \\ + \text{ isometries} \end{array} \right\} \to \mathsf{QM}. \tag{1}$$

Exercise 1.3. The ground states in Example 1.2 are (by definition) harmonic functions on M. For M closed, show that these comprise a vector space of dimension equal to $|\pi_0(M)|$. *Hint:* Using $\Delta = d^*d$, show $\Delta f = 0$ if and only if $df = 0$ for d the exterior derivative with adjoint d^*.

Example 1.4 (Adding a potential)**.** A common variation on the previous example adds the data of a *potential* $h \in C^\infty(M)$ for h bounded below. Keeping the same Hilbert space $\mathcal{H} = L^2(M)$, this potential modifies the Hamiltonian as $H = \Delta + h$. The ground states of this modified system are harder to pin down.

Definition 1.5. A quantum mechanical system $(\mathcal{H}, H) \in \mathsf{QM}$ is *compact* if $\mathrm{Tr}(e^{-tH}) < \infty$ for all $t > 0$.

Example 1.6. If M is closed (i.e., compact without boundary) the corresponding object of QM in Example 1.2 is compact [Dav89].

Convention 1.7. Unless stated otherwise, hereafter all quantum mechanical systems are compact, and all Hamiltonians are non-negative, $H \geq 0$.

Exercise 1.8 (Consequences of compactness)**.** Let $H \geq 0$ be a (self-adjoint, unbounded) Hamiltonian, and suppose that $\mathrm{Tr}(e^{-tH}) < \infty$ for every $t > 0$. Show that H has discrete spectrum, finite-dimensional eigenspaces, and no accumulation point in its spectrum, with the possible exception of the eigenvalue at ∞ (to be discussed further below). If $N(E)$ counts eigenvalues at most E with multiplicity, prove

$$N(E) \leq e^{tE}\,\mathrm{Tr}(e^{-tH}).$$

Hint: Use the spectral theorem to diagonalize e^{-tH}. If that doesn't suffice, see [HST10, §3].

1.2. **Symmetries, the renormalization group, and phases.**

Definition 1.9. A quantum mechanical system with *G-symmetry* is a functor $\mathbb{B}G \to \mathsf{QM}$.

To unpack this, recall that $\mathbb{B}G$ is the category with a single object with automorphisms G. Hence, the value of this functor on the unique object picks out $(\mathcal{H}, H) \in \mathsf{QM}$ together with a homomorphism $\rho\colon G \to \mathrm{PU}(\mathcal{H})$ such that $H = \rho(g)H\rho(g)^{-1}$ for all $g \in G$. In other words, we get a projective G-representation on $\mathcal{H}$ commuting with the Hamiltonian H. Equivalently, this is a central extension

$$1 \to U(1) \to \widehat{G} \to G \to 1$$

with a $\widehat{G}$-representation on $\mathcal{H}$ commuting with H [AS04].

Example 1.10. By functoriality of (1), the group of isometries of a Riemannian manifold M acts by symmetries on the quantum system $(L^2(M), \Delta)$. If we add a potential as in Example 1.4, the symmetry group typically gets smaller, corresponding to the subgroup of Riemannian isometries that preserve the potential h. This is a basic example of (explicit) *symmetry breaking*.

Remark 1.11. If G is a topological group, one should require the homomorphism $G \to \mathrm{PU}(\mathcal{H})$ to be continuous for the compact-open topology on $\mathrm{PU}(\mathcal{H})$. We will discuss these issues of topology shortly.

An important structure in quantum field theory is the renormalization group flow, but it can be hard to get ahold of mathematically. In quantum mechanics it is totally explicit.

Definition 1.12. Regard the positive real numbers $\mathbb{R}_{>0}$ as a group under multiplication. The *RG-flow* is the $\mathbb{R}_{>0}$-action on QM determined by $\lambda \cdot (\mathcal{H}, H) = (\mathcal{H}, \lambda H)$ for $\lambda \in \mathbb{R}_{>0}$.

In physics, topological theories have no time evolution. This leads to the following.

Definition 1.13. A quantum mechanical system is *topological* if $H = 0$.

Exercise 1.14. Show that a quantum mechanical system $(\mathcal{H}, H)$ is topological if and only if it is fixed under the RG-flow.

Example 1.15. For quantum mechanics on M, $(L^2(M), \Delta)$ is topological if and only if M is discrete.

Definition 1.16 (Aspirational)**.** The limits of the RG-flow for $\lambda \to 0$ and $\lambda \to \infty$ (when they exist) are called *UV-limit* and *IR-limit*, respectively, standing for *ultraviolet* (high energy) and *infrared* (low energy).

Relatedly, it is often important to know when a given quantum mechanical system can be deformed into another one.

Definition 1.17 (Aspirational)**.** Deformation classes of quantum systems are *phases*: two systems are in the same phase when they can be (continuously) deformed into each other.

To make sense out of Definitions 1.16 and 1.17, we will need to promote QM to a topological category. To supply some hope that deformations and limits are interesting, consider the following examples.

Example 1.18. Consider Example 1.4 for the family of potentials $s \cdot h \in C^\infty(M)$ for $s \in \mathbb{R}$ a parameter. Hence, at $s = 0$ there is no potential, while for $|s| \gg 0$ the potential dominates. It can be interesting to study the limit $s \to \infty$ of the Hamiltonian $H(s) = \Delta + s \cdot h$. A supersymmetric variation of such a limit is the subject of Exercise 1.40.

Example 1.19. Continuing Example 1.15 but supposing that $\dim(M) > 0$, the RG-flow is nontrivial. Then the IR-limit should be the topological theory determined by $\ker(\Delta) = \mathbb{C}^{\pi_0(M)}$: the nonzero eigenspaces get exponentially suppressed under $e^{-t\lambda\Delta}$ as $\lambda \to \infty$. In this sense the IR limit can be thought of as *zooming out*: we only see the connected components of M rather than any of the geometry captured by Δ. On the other hand, the UV-limit is $(L^2(M), 0)$, and this theory is not compact since $e^0 = \mathrm{id}_{L^2(M)}$ is not trace class.

Remark 1.20. The previous example is in some sense generic: we expect IR limits to exist, but UV limits often do not.

Remark 1.21. By our compactness assumption on quantum mechanical systems, $H = 0$ implies that $\mathcal{H}$ is finite-dimensional. This identifies (compact) topological quantum mechanical systems with finite-dimensional Hilbert spaces. This might sound familiar if you've thought about the 1-dimensional (unitary) cobordism hypothesis [Lur09a, Example 1.1.9].

Exercise 1.22 (The IR limit)**.** For a Hamiltonian H, let Π_0 be the orthogonal projection onto $\ker(H)$. Show that

$$\lim_{t\to\infty} e^{-tH} = \Pi_0$$

in the trace norm (and hence also in the operator norm). This gives one way to make precise sense of the IR limit of a (compact) quantum mechanical theory as the topological theory determined by its zero-energy states.

1.3. QM **as a topological category.** To promote QM to a topological category, we need a topology on the space of objects and the space of morphisms. As indicated by Remark 1.11, there is a candidate topology on the projective unitary operators, and Exercise 1.22 suggests a topology on the collection of Hamiltonians through a topology on the time-evolution operators e^{-tH}. This is all worked out carefully in [HST10, §2-5], building on [ST04, §3]. Below we will explain how to think about the resulting spaces.

It will be instructive to picture a Hamiltonian H in terms of its energy levels: we extract a sequence of real numbers $E_0, E_1, E_2, \dots$ and associated eigenspaces $\mathcal{H}_{E_i} \subset \mathcal{H}$. We organize this information as in Figure 1, where it is useful to represent a subspace $\mathcal{H}_{E_i}$ by a number of dots corresponding to its dimension. The properties of H deduced in Exercise 1.8 imply that the energy levels are discrete, and there can only be finitely many states (i.e., dots) occupying a given level.

Definition 1.23 ([HST10, Definition 3.2])**.** The collection $\{\mathcal{H}_{E_i}\}$ is a *configuration* on the nonnegative reals valued in finite-dimensional pairwise orthogonal subspaces of $\mathcal{H}$, see Figure 1. Endow the set $\mathrm{Conf}(\mathcal{H})$ of such configurations with the *generalized norm topology*.

To give a feel for this generalized norm topology, a deformation of a quantum mechanical system allows the energies to wiggle around, and also allows the subspaces $\mathcal{H}_{E_i} \subset \mathcal{H}$ to move around inside of $\mathcal{H}$. Energy levels can also split and merge, and this corresponds to direct sums of eigenspaces; schematically this looks like

$$\mathcal{H}_{E_i} \oplus \mathcal{H}_{E_j} \xrightarrow{\text{merge}} \mathcal{H}_E, \qquad \mathcal{H}_E \xrightarrow{\text{split}} \mathcal{H}_{E_i} \oplus \mathcal{H}_{E_j}, \qquad \mathcal{H}_E = \mathcal{H}_{E_i} \oplus \mathcal{H}_{E_j}.$$

In terms of Figure 1, when two energy levels collide we take the union of the dots, and energy levels may split by dividing the dots into subsets.

The RG-flow in Figure 1 is a simple rescaling of the vertical axis; explicitly

$$\mathcal{H}_E \xrightarrow{\mathrm{RG}_\lambda} \mathcal{H}_{\lambda E}, \qquad \lambda \in \mathbb{R}_{>0}$$

i.e., the subspaces of $\mathcal{H}$ are unchanged, but the energy labels E are rescaled by λ. The IR limit of the RG flow therefore sends all energy levels with $E_i \neq 0$ to infinity, while preserving the zero-energy states. On the other hand, the UV limit collapses all of the energy levels to the ground state. This limit only makes sense as a configuration if the initial configuration only had finitely many states (i.e., dots) to start.

Definition 1.24. Let $\mathsf{QM}(\mathcal{H})$ denote the topological category whose underlying category is the collection of quantum mechanical systems with underlying Hilbert space $\mathcal{H}$, where the object

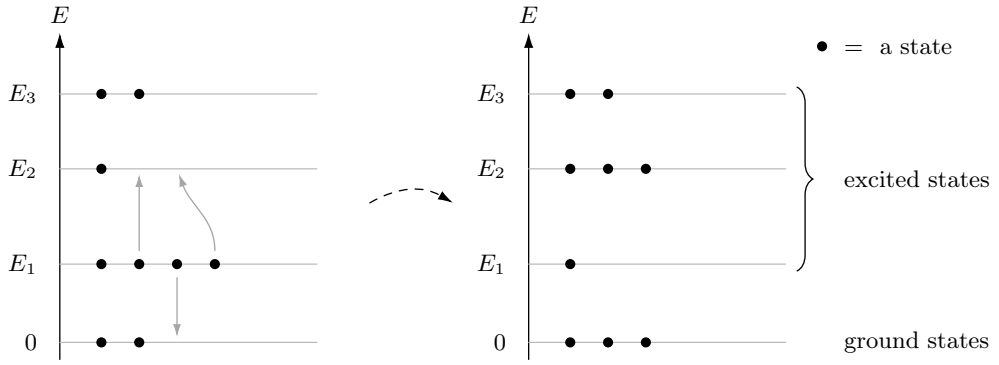


FIGURE 1. A picture of the energy levels of a Hamiltonian H, and a deformation in the space $\mathrm{Conf}(\mathcal{H})$. The horizontal lines correspond to eigenvalues E_i, and each dot • records a dimension of the corresponding E_i-eigenspace. Eigenspaces can split and merge, as indicated in moving from the picture on the left to the one on the right: two E_1-states move up to E_2 and another moves down to the ground state.

space is endowed with the generalized norm topology, and the morphisms are topologized via the compact open topology on $\mathrm{PU}(\mathcal{H})$.

Unwinding the above, $\mathsf{QM}(\mathcal{H})$ is a (topological) action groupoid

$$\mathsf{QM}(\mathcal{H}) \simeq \mathrm{Conf}(\mathcal{H}) /\!\!/ \mathrm{PU}(\mathcal{H})$$

for the action of the projective unitary group on the space of configurations. For the relevant topology on the unitary group, see [AS04, Proposition A1.1] and [EU14]; for continuity of the action see [BEG26]. If we fix a separable infinite-dimensional Hilbert space $\mathcal{H}$, a topological theory can be embedded in $\mathrm{Conf}(\mathcal{H})$ as a configuration supported at the zero eigenvalue labeled by a finite-dimensional subspace $V \subset \mathcal{H}$. The operator-theoretic interpretation is that the Hamiltonian is zero on V and is infinite on $V^\perp$: everything but the ground states have been pushed off to infinity. Equivalently, we allow for stabilizations of theories by adding arbitrary extra dimensions to the eigenspace at infinity. Such a perspective (or various equivalent repackagings) is crucial if we want to make sense of infinite limits of the RG flow.

Theorem 1.25 (Hohnhold-Stolz-Teichner)**.** *For any Hilbert space $\mathcal{H}$, the space* $\mathrm{Conf}(\mathcal{H})$ *is contractible. When $\mathcal{H}$ is an infinite-dimensional separable Hilbert space, we conclude that the geometric realization has the homotopy type* $|\mathsf{QM}(\mathcal{H})| \simeq K(\mathbb{Z}, 3)$.

Idea of the proof. For the first statement, perturb a given collection of Hamiltonians by δ so that their ground states have energy $> \epsilon$. Then the renormalization group flow gives a contracting homotopy of $\mathrm{Conf}(\mathcal{H})$ to the topological theory with $V = \{0\}$. For $\mathcal{H}$ an infinite-dimensional separable Hilbert space, this contractibility gives $|\mathsf{QM}(\mathcal{H})| \simeq |* /\!\!/ \mathrm{PU}(\mathcal{H})| \simeq B\,\mathrm{PU}(\mathcal{H}) \simeq K(\mathbb{Z}, 3)$, also using contractibility of the infinite-dimensional unitary group [AS04]. [1] □

Remark 1.26. For G a compact Lie group, one can repeat the above analysis for the category $\mathrm{Fun}(\mathbb{B}G, \mathsf{QM}(\mathcal{H}))$. The dots in Figure 1 are then labeled by an irreducible (projective) representation of G. These irreducible representations can merge and split, just as the dots did previously. The classifying space nearly has the same homotopy type as before, for essentially the same reasons: the zero representation provides a basepoint, and everything can be contracted to there. However, there is one new piece of data: a homomorphism $G \to \mathrm{PU}(\mathcal{H})$ classifying a central extension of G.

Remark 1.27. The projective unitary group is a 1-categorical shadow of a 2-categorical object: let $\widehat{\mathsf{QM}}$ denote the 2-groupoid whose objects are quantum mechanical systems, 1-morphisms are

[1] Since $U(\mathcal{H}) \simeq *$, the fiber sequence $U(1) \to U(\mathcal{H}) \to PU(\mathcal{H})$ implies that $PU(\mathcal{H})$ is a $K(\mathbb{Z}, 2)$.

unitary maps intertwining the Hamiltonians, and 2-morphisms are $U(1)$-scalars relating a pair of unitary maps. Then QM is gotten by identifying the 2-isomorphic 1-morphisms in $\widetilde{\mathsf{QM}}$.

1.4. **Supersymmetric quantum mechanics.**

Definition 1.28. A $\mathbb{Z}/2$-*graded Hilbert space* is a Hilbert space $\mathcal{H}$ with an orthogonal decomposition $\mathcal{H} = \mathcal{H}^{\text{even}} \oplus \mathcal{H}^{\text{odd}}$ where both summands $\mathcal{H}^{\text{even}}$ and $\mathcal{H}^{\text{odd}}$ are closed. The *fermion parity operator* $(-1)^F \colon \mathcal{H} \to \mathcal{H}$ acts by $+1$ on $\mathcal{H}^{\text{even}}$ and by -1 on $\mathcal{H}^{\text{odd}}$. When $\mathcal{H}$ is finite-dimensional, its *super dimension* is $\operatorname{sdim}(\mathcal{H}) = \operatorname{Tr}((-1)^F) = \dim \mathcal{H}^{\text{even}} - \dim \mathcal{H}^{\text{odd}}$.

Remark 1.29. Below we deliberately avoid lengthy discussions of sign conventions, and work with the above notion of $\mathbb{Z}/2$-graded Hilbert spaces as it communicates more transparently with standard linear algebra. We only note that there are superficially different conventions for $\mathbb{Z}/2$-graded Hilbert spaces that lead to equivalent categories, see Exercise 1.32. We recommend Luuk Stehouwer's thesis [Ste24] for a particularly clear-eyed view of the situation.

An operator $u \colon \mathcal{H} \to \mathcal{H}$ is *homogeneous* if it preserves the grading or swaps the grading, i.e.,

$$u = \begin{bmatrix} u_0 & 0 \\ 0 & u_1 \end{bmatrix}, \quad \begin{matrix} u_0 \colon \mathcal{H}^{\text{even}} \to \mathcal{H}^{\text{even}} \\ u_1 \colon \mathcal{H}^{\text{odd}} \to \mathcal{H}^{\text{odd}} \end{matrix} \quad \text{or} \quad u = \begin{bmatrix} 0 & v_1 \\ v_0 & 0 \end{bmatrix}, \quad \begin{matrix} v_0 \colon \mathcal{H}^{\text{even}} \to \mathcal{H}^{\text{odd}} \\ v_1 \colon \mathcal{H}^{\text{odd}} \to \mathcal{H}^{\text{even}} \end{matrix}$$

where the block form corresponds to the decomposition $\mathcal{H} = \mathcal{H}^{\text{even}} \oplus \mathcal{H}^{\text{odd}}$.

Definition 1.30. The *graded projective unitary group* of a $\mathbb{Z}/2$-graded Hilbert space is

$$\mathrm{PU}^{\pm}(\mathcal{H}) = \{u \in \mathrm{PU}(\mathcal{H}) \text{ homogeneous}\}$$

where $\mathrm{PU}^{+}(\mathcal{H})$ is the subgroup of grading-preserving projective unitary maps, while $\mathrm{PU}^{-}(\mathcal{H})$ is the component containing the grading-reversing projective unitary maps. We equip $\mathrm{PU}^{\pm}(\mathcal{H})$ with the compact-open topology.

Note $\mathrm{PU}^{-}(\mathcal{H}) \neq \emptyset$ if and only if $\mathcal{H}^{\text{even}}$ and $\mathcal{H}^{\text{odd}}$ are abstractly isomorphic. As such, given any $\mathbb{Z}/2$-graded Hilbert space $\mathcal{H}$, one can always choose an embedding $\mathcal{H} \hookrightarrow \mathcal{H}'$ where $\mathrm{PU}^{-}(\mathcal{H}') \neq \emptyset$. We adopt the following convention.

Convention 1.31. When a $\mathbb{Z}/2$-graded Hilbert space is infinite-dimensional, we assume that both $\mathcal{H}^{\text{even}}$ and $\mathcal{H}^{\text{odd}}$ are infinite-dimensional with countable bases.

Exercise 1.32. A *super Hermitian form* on a super vector space is a grading-preserving sesquilinear pairing

$$\langle -, - \rangle_s \colon \overline{\mathcal{H}} \hat{\otimes} \mathcal{H} \longrightarrow \mathbb{C}, \quad \overline{\langle v, w \rangle_s} = (-1)^{|v||w|} \langle w, v \rangle_s$$

where $\hat{\otimes}$ denotes the *graded* tensor product. The form is *positive* if it satisfies

$$i^{-|v|} \langle v, v \rangle_s \in \mathbb{R}_{\geq 0} \quad \text{and} \quad \langle v, v \rangle_s = 0 \iff v = 0.$$

Show that a positive super hermitian form determines an ordinary positive form on $\mathcal{H}$ whose even and odd subspaces are orthogonal. Then define a category of super Hilbert spaces that is equivalent to the category of $\mathbb{Z}/2$-graded Hilbert spaces and graded unitary maps.

Definition 1.33. Define the groupoid SQM of *quantum mechanical systems with supersymmetry* as having

- Objects: $\mathbb{Z}/2$-graded Hilbert spaces $\mathcal{H}$ equipped with an (unbounded) odd self-adjoint operator $Q \colon \mathcal{H} \to \mathcal{H}$.
- Morphisms: graded projective unitary operators $[u] \in \mathrm{PU}^{\pm}(\mathcal{H}, \mathcal{H}')$ with $Q' = (-1)^{|u|} u Q u^{-1}$.

The *Hamiltonian* of a supersymmetric system is $H := Q^2$. We continue to follow the compactness Convention 1.7, which requires $\operatorname{Tr}(e^{-tQ^2}) < \infty$ for all $t > 0$.

The configuration space picture from Figure 1 gets refined for supersymmetric systems, see Figure 2. The topology still allows eigenspaces to merge and split, but now there are restrictions coming from the fact that we have the odd square root Q of H: at each energy level there is an odd operator Q that squares to the energy. In particular, for positive-energy eigenspaces $E > 0$, the map $Q \colon \mathcal{H}_E^{\text{even}} \to \mathcal{H}_E^{\text{odd}}$ is an isomorphism.

Lemma 1.34. *There is a forgetful functor* $\mathsf{SQM} \to \mathsf{QM}$ *that on objects is* $(\mathcal{H}, Q) \mapsto (\mathcal{H}, Q^2)$.

Exercise 1.35. Prove the above: on objects forget the grading, and on morphisms verify that the graded projective unitary group includes into the ungraded one and the various actions are equivariant. Take care with the signs in the graded action!

The above forgetful functor does not have a section: given a quantum mechanical system $(\mathcal{H}, H)$, there need not exist a supersymmetric refinement in general.

Example 1.36 (Topological theories)**.** Topological theories have canonical supersymmetric refinements by taking $Q = 0$, i.e., $(\mathcal{H}, Q) = (\mathcal{H}, 0)$.

Example 1.37 (Differential forms)**.** Take M a closed oriented Riemannian manifold and $\mathcal{H}$ the L^2-completion of $\Omega^\bullet(M)$, i.e., the $\mathbb{Z}/2$-graded vector space of square-integrable differential forms. The Hodge-de Rham operator $Q = d + d^*$ defines a (compact) supersymmetric quantum mechanical system.

Remark 1.38. In the previous example, we tacitly are presenting a Hilbert space as a vector space together with an inner product (C^∞-differential forms and the Hodge inner product). We will continue to make such identifications in geometric examples below.

Remark 1.39. Definition 1.33 encodes quantum mechanical systems with $\mathcal{N} = 1$ supersymmetry, meaning a single odd square root Q of H. One can ask for any number $\mathcal{N}$ of (super) commuting odd square roots,

$$H = Q_i^2 = \frac{1}{2}[Q_i, Q_i], \qquad [Q_i, Q_j] = Q_iQ_j + Q_jQ_i = 0,\ i \neq j.$$

The previous Example 1.37 naturally has $\mathcal{N} = 2$, with second supersymmetry specified by $Q' = i(d - d^*)$. Witten studies this system in detail [Wit82]. We give a taste in the next exercise and subsequent remark.

Exercise 1.40 (Witten deformation)**.** We modify Example 1.37 using the data of a function $h \in C^\infty(M)$: take

$$d_s = e^{-sh} d\, e^{sh} \qquad Q_s = d_s + d_s^*, \qquad \Delta_s = Q_s^2.$$

Show the d_s-cohomology is independent of s and hence naturally isomorphic to $H^\bullet_{\mathrm{dR}}(M)$. *Hint:* Consider the map $\Omega^\bullet(M) \to \Omega^\bullet(M)$ given by $\omega \mapsto e^{sh}\omega$.

Remark 1.41. Witten analyzes the kernel of Δ_s in the limit $s \to \infty$ when h is a Morse function. He shows that the super dimension of this kernel is a signed count of the critical points of h, and goes on to explain how the entire Morse complex (and the Morse inequalities) can be extracted from understanding the low-lying eigenvalues of Δ_s in the large s limit. By now Witten's arguments are all rigorous, e.g., see [HS85].

Example 1.42 (Spin manifolds)**.** Let M be a closed even-dimensional Riemannian spin manifold and $\mathcal{H} = L^2(M; \mathbb{S})$ the L^2-sections of the spinor bundle $\mathbb{S} \to M$. Taking $Q = \not{D}$ the Dirac operator gives a $\mathcal{N} = 1$ supersymmetric quantum mechanical system. More generally, for $E \to M$ a vector bundle with hermitian metric and metric connection, one can take $\mathcal{H} = L^2(M; \mathbb{S} \otimes E)$ with $Q = \not{D}_E$ the twisted Dirac operator.

Remark 1.43. There is some ambiguity in the term "spinor bundle", which results in different flavors of the previous two examples. We mention two variants: (i) if M is even-dimensional, take the usual $\mathbb{Z}/2$-graded spinor bundle; and (ii) for M of arbitrary dimension, take the Clifford-linear Dirac operator, viewing the corresponding spinors with their Cl_n-action as a $\mathbb{Z}/2$-graded bundle. We will tacitly work with Clifford-linear Dirac operators throughout. The index of these operators is defined to be the graded Clifford module $[\ker(\not{D})] \in K^n(\mathrm{pt})$. This is particularly relevant for Real K-theory; we refer to [LM89] for detailed explanations.

As in the quantum mechanics examples, let $\mathsf{SQM}(\mathcal{H})$ denote supersymmetric quantum mechanical systems on a fixed separable infinite-dimensional Hilbert space $\mathcal{H}$. The *Witten index* is the assignment

$$\mathsf{SQM}(\mathcal{H}) \to \mathbb{Z}, \qquad Q \mapsto \mathrm{sdim}(\ker(Q))$$

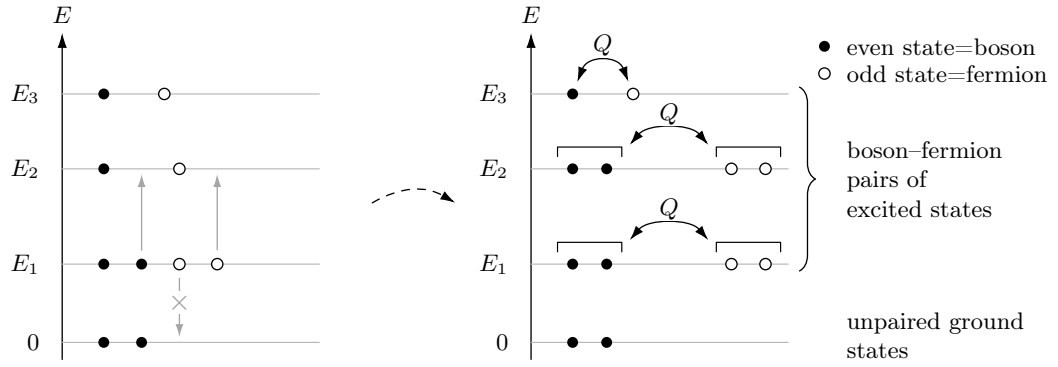


FIGURE 2. A picture of the energy levels of a Hamiltonian H with specified supersymmetry Q satisfying $H = Q^2$. The states with energy $E > 0$ come in pairs that are interchanged by Q. Eigenspaces can still split and merge, but now are constrained by the existence of Q, as indicated by the forbidden single fermion moving from an excited state into the ground state.

sending a supersymmetric quantum mechanical system to the super dimension of its ground states. Note that odd morphisms reverse the sign of the Witten index.

Remark 1.44. The McKean–Singer formula [MS67] identifies the Witten index with the expression

$$\operatorname{sdim}(\ker(Q)) = \operatorname{Tr}\left((-1)^F e^{-tQ^2}\right)$$

often called the (Ramond) *partition function* of the theory.

Exercise 1.45 (Supersymmetric cancelation)**.** Let Q be odd and self-adjoint, with e^{-tQ^2} trace class. Using that Q gives an odd isomorphism between the even and odd parts of every positive eigenspace of Q^2 (Figure 2), deduce that

$$\operatorname{sTr}(e^{-tQ^2}) = \operatorname{sdim}\ker(Q) = \operatorname{ind}\big(Q\colon \mathcal{H}^{\text{even}} \to \mathcal{H}^{\text{odd}}\big).$$

In particular, the partition function is independent of t.

Theorem 1.46 ([HST10])**.** *For an infinite-dimensional separable $\mathbb{Z}/2$-graded Hilbert space $\mathcal{H}$ satisfying the convention above, we have*

$$\operatorname{sConf}(\mathcal{H}) \simeq BU \times \mathbb{Z}$$

Consequently, $\mathsf{SQM} = \operatorname{sConf}(\mathcal{H}) /\!\!/ \operatorname{PU}^{\pm}(\mathcal{H})$ *is a classifying stack for twisted equivariant K-theory. Under this equivalence, the Witten index is the Atiyah–Singer index that reads off the connected component* $\operatorname{Ind}\colon BU \times \mathbb{Z} \to \mathbb{Z}$.

Remark 1.47. The unbounded operator model $\operatorname{sConf}(\mathcal{H}) /\!\!/ \operatorname{PU}^{\pm}(\mathcal{H})$ for twisted equivariant K-theory is constructed in [BEG26], closely following ideas of Freed–Hopkins–Teleman [FHT11, §3]. Here, $\operatorname{PU}^{+}(\mathcal{H})$ encodes twists valued in $H^3(-;\mathbb{Z})$, while the grading homomorphism $\operatorname{PU}^{\pm}(\mathcal{H}) \to \mathbb{Z}/2$ encodes twists valued in $H^1(-;\mathbb{Z}/2)$. Equivalently, $\operatorname{PU}^{\pm}(\mathcal{H})$ has the homotopy type of the 2-truncation of $\mathrm{GL}_1(\mathrm{KU})$. Twisted *equivariant* K-theory arises in this context by considering homotopy classes of maps of stacks $[X/G] \to [\operatorname{sConf}(\mathcal{H}) /\!\!/ \operatorname{PU}^{\pm}(\mathcal{H})]$ for a G-space X.

1.5. **Renormalization group flow categories.** One useful way to think about the space of quantum mechanical systems (and more generally, spaces of quantum field theories) is via flow categories. We review the general setup first.

Definition 1.48. For a space X equipped with a flow $\mathbb{R}_{>0} \times X \to X$, the *flow category* Flow_X has

- Objects: $\mathbb{R}_{>0}$-fixed points in X.
- Morphisms: (broken) flow lines between fixed points.

We regard Flow_X as a topological category for the natural topology on its objects and morphisms.

Theorem 1.49 (Cohen–Jones–Segal [CJS95]; Calle–Liu [CL26])**.** *For X a closed Riemannian manifold equipped with the gradient flow of a (tame) Morse function, there is a homotopy equivalence*

$$|\mathrm{Flow}_X| \simeq X.$$

In quantum field theory, there is a collection of results called *Zamolodchikov c-Theorems*. The original result due to Zamolodchikov produces a function called c on 2-dimensional (renormalizable) theories that decreases monotonically along RG flow lines [Zam86]. This has led to a lot of useful thinking by viewing the renormalization group flow as a gradient flow.

Problem 1 $(\star\star\star)$**.** Formulate and prove Zamolodchikov's c-Theorem in a mathematically rigorous framework of 2-dimensional field theory, e.g., the bordism framework in the next lecture.

The c-Theorem gives a starting point for how to visualize spaces of quantum field theories following the analogy with Morse theory. Conformal field theories are essential building blocks, amounting to specifying the collection of "handles". The index of the handle has attendant terminology in physics: descending directions correspond to *relevant* operators, ascending directions are *irrelevant* operators, and operators that stay within a critical submanifold (as for a Morse–Bott function) are (exactly) *marginal* operators. The ways these handles are attached and flow lines compose carry important physical content, including various *dualities* of quantum field theories [Sei95].

To get a basic feeling for how these renormalization group flow categories work, we again turn to quantum mechanics as the guiding example: we will apply the version of RG flow from Definition 1.12 to the topological categories

$$\widetilde{\mathsf{QM}} := [\mathrm{Conf}(\mathcal{H}) /\!\!/ \mathrm{U}(\mathcal{H})], \qquad \widetilde{\mathsf{SQM}} := [\mathrm{sConf}(\mathcal{H}) /\!\!/ \mathrm{U}^+(\mathcal{H})]$$

which are non-projective versions of the ones studied above; the projective case is also manageable, but more complicated. Furthermore, RG flow on $\widetilde{\mathsf{SQM}}$ turns out to recover some beautiful mathematics from algebraic K-theory in a rather unexpected way. This type of structure makes it feel inevitable that flow categories for higher-dimensional quantum field theories will encode very rich mathematics.

By Exercise 1.14, the RG fixed points in $\widetilde{\mathsf{QM}}(\mathcal{H})$ are topological theories. We identify the space of such topological theories with embeddings $V \subset \mathcal{H}$ of a finite-dimensional inner product space.

Construction 1.50. *The flow category* $\mathrm{Flow}_{\widetilde{\mathsf{QM}}}$ *is a topological category whose objects are embeddings $V \subset \mathcal{H}$ of a finite-dimensional inner product space, and morphisms are pairs $u \in \mathrm{U}(\mathcal{H})$ and an isometric embedding $u(V) \hookrightarrow W$ together with a positive self-adjoint operator $H_\perp \colon V^\perp \to V^\perp$, where $V^\perp$ denotes the orthogonal complement of $u(V)$ in W.*

In other words, a flow line in $\mathrm{Hom}(V, W)$ is a Hamiltonian on W of the form $H_W = 0_V \oplus H_\perp$ for $H_\perp$ positive. The identity flow line corresponds to the identity embedding $V \subset V$ where $V^\perp = \{0\}$ has a unique invertible (and vacuously positive) operator.

Proposition 1.51. *The classifying space* $|\mathrm{Flow}_{\widetilde{\mathsf{QM}}}|$ *is contractible.*

Proof. Observe that $\mathrm{Flow}_{\widetilde{\mathsf{QM}}}$ has a weakly initial object $\{0\}$: $\mathrm{Hom}(\{0\}, V)$ is the Cartesian product of $\mathrm{U}(\mathcal{H})$ with the space of positive operators on V, and hence contractible (using Kuiper's theorem [Kui65]). This implies the flow category of $\mathrm{Conf}(\mathcal{H})$ is contractible. □

We repeat the same story but for supersymmetric quantum mechanics.

Construction 1.52. *The flow category* $\mathrm{Flow}_{\widetilde{\mathsf{QM}}}$ *is the topological category with objects graded embeddings $V \subset \mathcal{H}$ of a finite-dimensional $\mathbb{Z}/2$-graded Hilbert space, and morphisms are $u \in \mathrm{U}^+(\mathcal{H})$ with a grading-preserving isometric embedding $u(V) \hookrightarrow W$ together with an odd, invertible self-adjoint operator $Q_\perp \colon V^\perp \to V^\perp$, where $V^\perp$ denotes the orthogonal complement of $u(V)$ in W.*

Exercise 1.53. Identifying finite-dimensional super Hilbert spaces V and W with their corresponding topological field theories, show

$$\begin{array}{c}\text{There exists a supersymmetric}\\ \text{flow line between } V \text{ and } W\end{array} \iff \mathrm{sdim}(V) = \mathrm{sdim}(W) \iff [V] = [W] \in \mathrm{K}^0(\mathrm{pt}).$$

From the above exercise, we conclude $\pi_0(|\mathrm{Flow}_{\widetilde{\mathsf{SQM}}}|) = \mathbb{Z}$. In particular this flow space is not contractible. One can further compute the morphism spaces explicitly; for example

$$\mathrm{Hom}(\mathbb{C}^{0|0}, \mathbb{C}^{n|n}) \simeq \mathrm{GL}_n(\mathbb{C}),$$

where any invertible odd self-adjoint operator $Q\colon \mathbb{C}^{n|n} \to \mathbb{C}^{n|n}$ has the form

$$Q = \begin{bmatrix} 0 & A \\ A^* & 0 \end{bmatrix}, \quad A \in \mathrm{GL}_n(\mathbb{C}).$$

Composing flow lines corresponds to the block sum

$$\mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_m(\mathbb{C}) \to \mathrm{GL}_{n+m}(\mathbb{C}).$$

From this one finds that supersymmetric flow lines construct the group completion of 1-dimensional topological theories.

Theorem 1.54 (Quillen)**.** *The geometric realization of the flow category for supersymmetric quantum mechanics is* $|\widetilde{\mathsf{SQM}}| \simeq \mathrm{BU} \times \mathbb{Z}$*. The inclusion of purely even topological field theories into the flow category*

$$\coprod_{k\geq 0} [\mathrm{pt} \,/\, \mathrm{U}(k)] \hookrightarrow \mathrm{Flow}_{\widetilde{\mathsf{SQM}}}$$

geometrically realizes to the group completion map.

Proof sketch. Spelling out the geometric realization $|\mathrm{Flow}_{\widetilde{\mathsf{SQM}}}|$, one obtains a topological version of Quillen's $S^{-1}S$ construction that provides a classifying space model for the group completion of finite-dimensional vector spaces [HST10, §7]. □

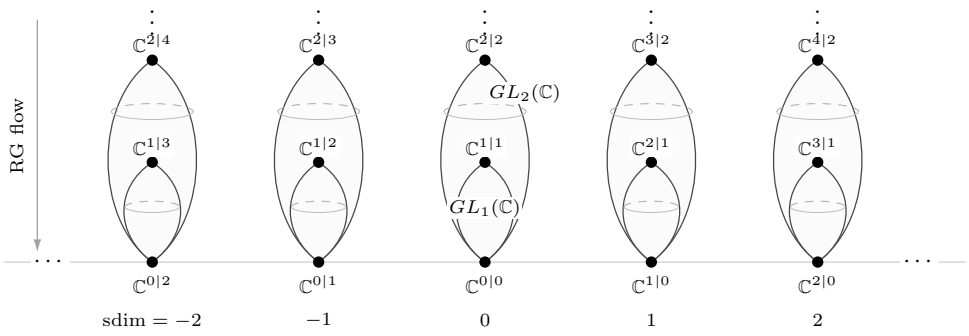


FIGURE 3. The flow category for $\widetilde{\mathsf{SQM}}$. The dots • represent topological theories determined by a super vector space $\mathbb{C}^{n|m}$, where flow lines exist between topological theories if and only if the super vector spaces have the same super dimension. Over $\mathbb{C}^{0|0}$, the spaces of flow lines are the nested inclusions $\mathrm{GL}_1(\mathbb{C}) \subset \mathrm{GL}_2(\mathbb{C}) \subset \ldots$. The other spaces of flow lines are Stiefel manifolds.

Much of K-theory can then be recast in terms of supersymmetric flow lines between topological theories.

Example 1.55 (The Bott class as supersymmetric deformations of a topological theory)**.** The theorem above gives a way to understand K-theory classes in terms of supersymmetric deformations of topological theories. We will explain how this goes for the Bott class. Take the map $\mathbb{C} \setminus \{0\} \to \mathrm{Hom}(\{0\}, \Lambda\mathbb{C})$ that sends $v \in \mathbb{C}$ to $\iota_v + \wedge v\colon \Lambda\mathbb{C} \to \Lambda\mathbb{C}$ – its square is the positive operator $|v|^2 \cdot \mathrm{id}$ on the complement $\{0\}^\perp$. In the limit $v \to 0$, we obtain the identity element $\mathrm{Hom}(\Lambda\mathbb{C}, \Lambda\mathbb{C})$, i.e., the unique (and vacuously invertible) operator on $\{0\} \subset \Lambda\mathbb{C}$. This

family of operators over $\mathbb{C}$ gives a (real) 2-dimensional family of supersymmetric deformations of the topological theory $\Lambda\mathbb{C}$. In K-theory, this family represents the Bott class [HST10, §1] $\beta \in ku^0(\mathbb{C}, \mathbb{C} \setminus \{0\})$.

Exercise 1.56 (The Bott family)**.** For $v \in \mathbb{C}$, let

$$Q_v = \iota_v + v \wedge -\colon \Lambda^\bullet \mathbb{C} \longrightarrow \Lambda^\bullet \mathbb{C}.$$

Using the contraction identity, show that

$$Q_v^2 = |v|^2 \,\mathrm{id}\,.$$

Conclude that Q_v is invertible for $v \neq 0$, and explain why the family $\{Q_v\}_{v\in\mathbb{C}}$ determines a relative K-theory class in

$$K^0(\mathbb{C}, \mathbb{C} \setminus \{0\}).$$

2. Quantum field theories as representations of bordism categories

On both the physics and the mathematics side, the quest continues to figure out what quantum field theory really *is*. There are now several approaches that each capture some piece of what is surely a much larger idea. Most of these mathematical frameworks have something useful to say about Conjecture A. Partly because of its historical relevance and partly for the geometric intuition it imparts, in these lectures we will follow Graeme Segal's idea [Seg04]: quantum field theories are representations of bordism categories. We note that Atiyah adopted this idea in the case of topological quantum field theories [Ati88], which is the better-studied variant of this definition.

2.1. A definition of quantum field theory.

Definition 2.1 (Sketch)**.** A *d-dimensional quantum field theory with geometry $\mathcal{F}$* is a symmetric monoidal functor

$$Z\colon \left(\mathrm{Bord}_d(\mathcal{F}), \coprod\right) \to (\mathrm{Vect}, \otimes) \tag{2}$$

where $\mathrm{Bord}_d(\mathcal{F})$ is the bordism category of oriented $(d-1)$-dimensional manifolds with geometry $\mathcal{F}$.

Our primary goal today will be to connect Definition 2.1 with the previous lecture.

Example 2.2 (Quantum mechanics)**.** Quantum mechanics is a 1-dimensional quantum field theory, where the world-line of the particle is the 1-dimensional cobordism.

Before pursuing the above example in detail, we comment on general features of Definition 2.1. We visualize a quantum field theory as in Figure 4: a bordism Σ with additional geometric decorations is assigned a linear map between the vector spaces assigned to its boundary. The geometry $\mathcal{F}$ is specified as a sheaf or stack on the site of d-dimensional manifolds. Important examples include Riemannian metrics, the stack of G-bundles with connection (gauge fields), or maps into a smooth manifold (scalar fields). We use the notation $\mathsf{QFT}_d(\mathcal{F})$ for the groupoid whose objects are symmetric monoidal functors (2) and whose morphisms are natural isomorphisms. We take the simplified notation $\mathsf{TFT}_d := \mathsf{QFT}_d(\mathrm{pt})$ for d-dimensional topological field theories, CFT_d for d-dimensional conformal field theories, and $\mathsf{QFT}_d := \mathsf{QFT}_d(\mathrm{Riem})$ for d-dimensional Riemannian theories.

In Definition 2.1, the target category Vect fixes the functional analytic context: one decides on the flavor of topological vector spaces, the continuous maps between them, and the (usually completed) tensor product of these vector spaces. The choices are usually dictated by the class of quantum field theories of interest.

We emphasize that Definition 2.1 is woefully incomplete as stated. There are additional features one must add to complete the definition, and optional features that are potentially desirable depending on the context. The experts continue to debate these technicalities. To give a flavor for this ongoing discussion, we indicate some of these features.

- The boundaries of bordisms must be collared if composition is to be defined via gluing. There is a variant of Definition 2.1 that encodes cutting axioms rather than gluing ones, placing the collar data further in the background (and in some ways avoids it).

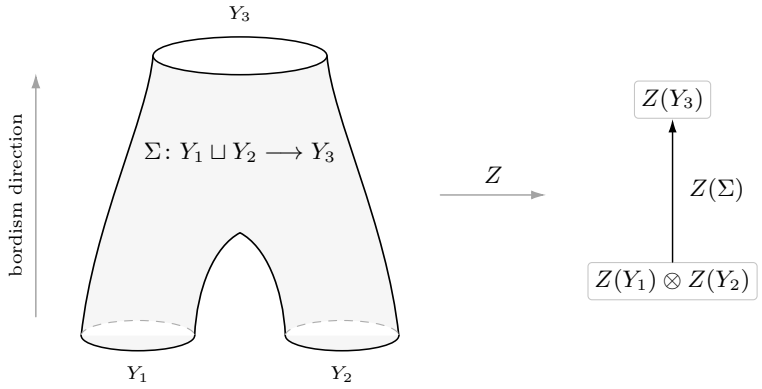


FIGURE 4. The value of a field theory Z on a bordism $\Sigma\colon Y_1 \coprod Y_2 \to Y_3$. The surface Σ can be equipped with a geometry, e.g., a Riemannian metric, in which case the value of Z will depend on that geometry. The boundary components Y_i are also typically decorated with (a germ) of the same geometry. We take the "positive" convention here and throughout: pictures of bordisms have source at the bottom and target at the top.

- The objects of $\mathrm{Bord}_d(\mathcal{F})$ might themselves have geometric structures with some relation to the bulk, e.g., we might ask for boundaries of Riemannian manifolds to be cylindrical or geodesic. Often, they are taken to be germs of d-manifolds around a $d-1$-manifold.
- The functor Z should depend continuously or smoothly on input geometric parameters, e.g., on the metric if $\mathcal{F}$ is the sheaf of Riemannian metrics.
- The objects and morphisms in $\mathrm{Bord}_d(\mathcal{F})$ can themselves have internal automorphisms, e.g., G-bundle automorphisms if $\mathcal{F}$ is the stack of G-bundles.
- The units in the category $\mathrm{Bord}_d(\mathcal{F})$ typically involve degenerate bordisms, e.g., Riemannian d-manifolds of zero volume. It is also possible to define $\mathrm{Bord}_d(\mathcal{F})$ to *not* be unital. (cf. Theo Johnson-Freyd's lecture course)
- Supersymmetry is implemented by taking *supermanifolds* as bordisms [ST11]. Equipping these with geometric structures (e.g., metrics) is a subtle task.

One nice feature of the cobordism framework is that many structures in quantum field theory are inherited from naturality in the sheaf parameter $\mathcal{F}$. Here are three examples of this.

Example 2.3 (Topological theories)**.** For any sheaf $\mathcal{F}$, the terminal map $\mathcal{F} \to *$ gives a forgetful functor

$$\mathrm{Bord}_d(\mathcal{F}) \to \mathrm{Bord}_d \tag{3}$$

to the topological bordism category. This induces a map on field theories

$$\mathsf{TFT}_d \to \mathsf{QFT}_d(\mathcal{F})$$

whose image consists of quantum field theories that are topological in the sense that they do not depend on the geometry $\mathcal{F}$.

Example 2.4 (Background G-symmetry)**.** As a special case of Example 2.3, the forgetful map $\mathrm{Bord}_d(\mathbb{B}^\nabla G) \to \mathrm{Bord}_d$ forgets the data of a G-bundle with connection. A lift along the induced map on field theories equips a topological field theory with *background G-symmetry*. The same applies for maps induced by a projection $\mathbb{B}^\nabla G \times \mathcal{F} \to \mathcal{F}$. We will investigate this structure further in §4.

Example 2.5 (The renormalization group flow)**.** Set $\mathcal{F} = \mathrm{Riem}$ to be the sheaf of Riemannian metrics. There is an $\mathbb{R}_{>0}$-action on $\mathcal{F}$ that dilates the size of metrics by $\lambda \in \mathbb{R}_{>0}$. This induces an $\mathbb{R}_{>0}$-action on $\mathrm{Bord}_d(\mathcal{F})$, which in turn induces an $\mathbb{R}_{>0}$-action on QFT_d—the renormalization group flow.

Remark 2.6. Fixed points for the renormalization group flow of Example 2.5 are field theories invariant under *global* rescalings. This is a priori distinct from being a conformal field theory,

which demands invariance under all *local* rescalings. There is an evident inclusion

$$\mathsf{CFT}_d \hookrightarrow \mathsf{QFT}_d(\text{scale invariant}) \tag{4}$$

and hence one can ask whether this inclusion is an equivalence of groupoids. Following an argument of Zamolodchikov, Polchinski argues (at a physics level of rigor) that theories in a suitable class of two-dimensional local scale-invariant quantum field theories are conformally invariant [Pol88].

Problem 2 ($\star\star\star$)**.** Adapt the Zamolodchikov–Polchinski argument to find hypotheses under which (4) is an equivalence.

Polchinski's argument requires a well-defined stress tensor, which gives a warm-up problem.

Problem 3 ($\star$)**.** Define stress-energy tensors in this framework, expanding on [Seg04, §9].

2.2. **Reflection Positivity.** Definition 2.1 by itself does not lead to Hilbert structures and self-adjoint Hamiltonians in quantum mechanics. Rather, we require an additional layer called *reflection positivity* [OS73, FH21]. This is the Wick-rotated (or Euclidean) analogue of unitarity in quantum field theory. Not all quantum theories of interest in mathematical physics have this feature. Notably, topological twists of supersymmetric theories are generally not reflection positive. But the theories relevant to Conjecture A carry this additional structure, making this an important ingredient below.

Definition 2.7. A *reflection structure* for a field theory

$$Z\colon \mathrm{Bord}_d \to \mathrm{Vect} \tag{5}$$

is a C_2-equivariance datum relative to the actions generated by orientation reversal of bordisms and reversing the complex structure on vector spaces. For a geometric bordism category, reflection positivity requires input data of a C_2-action on $\mathrm{Bord}_d(\mathcal{F})$ that covers orientation reversal under (3).

Using bar notation to denote both orientation reversal (when applied to a manifold) and reversal of complex structure (when applied to a vector space) C_2-equivariance data are isomorphisms

$$\overline{Z(Y)} \cong Z(\overline{Y}) \tag{6}$$

for each object $Y \in \mathrm{Bord}_d$, with compatibility properties with morphisms (i.e., bordisms). Equivalently, (6) is data for Z being a *homotopy C_2-fixed point* for the diagonal C_2-action on functors.

Definition 2.8. A reflection structure is *reflection positive* if the hermitian pairings induced by (6)

$$\overline{Z(Y)} \otimes Z(Y) \cong Z(\overline{Y}) \otimes Z(Y) \to \mathbb{C}\,.$$

are positive definite.

There are many possible elaborations of reflection positivity. An important one for us comes when combining the above C_2-equivariance with spin structures: the category of spin manifolds has BC_2-action, i.e., every spin manifold has a canonical automorphism which "flips the sheets" of its spin frame bundle. Similarly, the category sVect of super vector spaces has a BC_2-action that is the parity involution $(-1)^F$ on a super vector space.

Definition 2.9. A field theory

$$\mathrm{Bord}_d^{\mathrm{Spin}} \to \mathrm{sVect}$$

satisfies spin statistics if Z is BC_2-equivariant for the actions described above. For brevity, we will often refer to this condition as a *spin theory*. For a geometric bordism category, we formulate an analogous definition relative to a choice of BC_2-action on $\mathrm{Bord}_d^{\mathrm{Spin}}(\mathcal{F})$ covering the BC_2-action on spin bordisms without $\mathcal{F}$-structure.

Problem 4 ($\star$)**.** Give a general formulation of reflection positivity for geometric bordism categories.

2.3. **Unpacking the definition in dimension 1.** We will now unpack the definition of a 1-dimensional quantum field theory, starting with the most basic structure of a 1-dimensional topological theory, and gradually building our way up to the example of supersymmetric quantum mechanics as a representation of a bordism category.

The values of a 1-dimensional topological theory on connected objects (the point with its two orientations) and connected morphisms (the circle or the interval equipped with an orientation and source and target data) are given in Figure 5.

$\mathrm{Bord}_1^{\mathrm{or}}$	$\longrightarrow$	Vect
pt_+	$\longmapsto$	V_+
pt_-	$\longmapsto$	$V_- \cong V_+^\vee$
$\overset{\pm}{\underset{\pm}{\downarrow}}$	$\longmapsto$	$\mathrm{id}_{V_\pm}$
$\overset{\pm\ \mp}{\frown}$	$\longmapsto$	$V_\pm \otimes V_\mp \to \mathbb{C}$
$\overset{\pm\ \mp}{\smile}$	$\longmapsto$	$\mathbb{C} \to V_\pm \otimes V_\mp$
$\bigcirc$	$\longmapsto$	$\mathrm{Tr}(\mathrm{id}_{V_\pm})$

FIGURE 5. The values of a 1-dimensional oriented topological field theory.

The above is a redundant list: Zorro's lemma shows that V_+ and V_- are dual to each other, and hence the entire theory is determined by its value $Z(\mathrm{pt}_+)$. Duality further requires $Z(\mathrm{pt}_+)$ be finite-dimensional. A reflection structure supplies isomorphisms $V_\pm \simeq \overline{V}_\mp$, and the positivity property demands that the hermitian pairing is positive-definite:

$$\overline{V} \otimes V \cong V_- \otimes V_+ \to \mathbb{C}, \qquad V := V_+ \simeq \overline{V}_-.$$

Adding spin structures to the bordisms and $\mathbb{Z}/2$-gradings to the vector spaces enhances the list in Figure 5 in the following way. There are still only two isomorphism classes of objects, denoted $\mathrm{pt}_\pm$, but now they each have a $\mathbb{Z}/2 \simeq \mathrm{Spin}(0)$-automorphism coming from the automorphism of the spin principal bundle. A *spin theory* (as in Definition 2.9) sends this $\mathbb{Z}/2$-automorphism to the parity involution $(-1)^F$ on a $\mathbb{Z}/2$-graded vector space. The intervals can be pre- and post-composed with this automorphism, and so the resulting identity map, pairing, and copairing get composed with $(-1)^F$. Gluing the spin interval to a spin circle in the two possible ways results in two spin circles: the bounding and the nonbounding circle. From the above discussion, the value of a spin theory on these circles is

$\mathrm{Bord}_1^{\mathrm{Spin}}$	$\longrightarrow$	sVect
$\bigcirc^{ap}$	$\longmapsto$	$\dim(V) = \mathrm{Tr}(\mathrm{id}_V)$
$\bigcirc^{p}$	$\longmapsto$	$\mathrm{sdim}(V) = \mathrm{Tr}((-1)^F).$

corresponding to the dimension and super dimension of the value on the point. Above, $\bigcirc^p$ denotes the circle with periodic (or non-bounding, odd, Ramond) spin structure, while $\bigcirc^{ap}$ carries the antiperiodic (or bounding, even, Neveu–Schwarz) spin structure.

In the next layer of generalization we leave the topological world behind and consider Riemannian field theories. As 1-dimensional connected Riemannian manifolds are characterized by their length, the assignments in Figure 5 are augmented as in Figure 6. In particular, we have intervals

$$I_t = \overset{+}{\underset{+}{\overset{\downarrow t}{}}}, \quad I_{t_1} \circ I_{t_2} \simeq I_{t_1+t_2}$$

with length $t \in \mathbb{R}_{\geq 0}$, where composition in the bordism category corresponds to addition in the semigroup $(\mathbb{R}_{\geq 0}, +)$. Therefore Z determines a representation of this semigroup $Z(I_t)\colon V \to V$.

Reflection positivity implies that this representation takes values in self-adjoint operators. In good analytical situations (see Exercise 2.13), such a representation is determined by a self-adjoint infinitesimal generator H,

$$\exp(-tH) = Z(I_t)\colon V \to V. \tag{7}$$

In contrast to the topological case, the value of Z on the point no longer determines the theory: dualizability is lost in the generalization to geometric bordisms. The remnant of this structure is the fact that $Z(I_t)$ is trace class, which imposes a kind of Ind-finiteness condition on the Hamiltonian: H has finite-dimensional eigenspaces and spectrum bounded below.

$\mathrm{Bord}_1^{\mathrm{Spin}}(\mathrm{Riem})$	$\longrightarrow$	sVect
pt_+	$\longmapsto$	V
	$\longmapsto$	$e^{-tH}\colon V \to V$
	$\longmapsto$	$\langle -, e^{-tH} - \rangle\colon \overline{V} \otimes V \to \mathbb{C}$
	$\longmapsto$	$\mathbb{C} \to V \otimes \overline{V},\ 1 \mapsto e^{-tH}$
	$\longmapsto$	$\mathrm{Tr}(e^{-tH}) \in C^\infty(\mathbb{R}_{>0})$
	$\longmapsto$	$\mathrm{sTr}(e^{-tH}) \in C^\infty(\mathbb{R}_{>0})$

FIGURE 6. The values of a 1-dimensional, spin, reflection positive Riemannian field theory. In particular, the values on the bordisms with the opposite orientation are determined by the ones above by taking opposite complex structures, and the spin flip automorphism coincides with the grading involution.

Example 2.10 (Quantum mechanics)**.** For M a closed Riemannian manifold, define a reflection positive Riemannian field theory

$$\mathrm{Bord}_1(\mathrm{Riem}) \to \mathrm{Vect}$$

via the data in Figure 6 by

$$V = C^\infty(M), \quad H = \Delta, \quad \langle f, g\rangle = \int \overline{f} g \, d\mathrm{vol}$$

where Δ is the Laplacian and $d\mathrm{vol}$ is the Riemannian volume form on M. This leads to a semigroup representation (7) whose trace (partition function) encodes lots of interesting geometric information about M [Kac66].

Remark 2.11. Especially when working with unital bordism categories, quantum field theories in the sense of Definition 2.1 tend to give pre-Hilbert spaces, i.e., a topological vector space with an inner product that is not necessarily complete relative to the induced topology. In the previous example, we use the data of $V = C^\infty(M)$ and the inner product $\langle -, -\rangle$ to construct the Hilbert space $L^2(M)$. The domain of the operator Δ (and its powers) acting on $L^2(M)$ can also be used to construct the embedding $C^\infty(M) \subset L^2(M)$, so this is really a question of how one presents a quantum mechanical system rather than a shift in the framework.

Remark 2.12. Example 2.10 is not presented above as a spin theory. It can be made into one in a trivial way by regarding V as a $\mathbb{Z}/2$-graded vector space in purely even degree.

Exercise 2.13 (The infinitesimal generator)**.** Let $T_t = e^{-tH}$ for a nonnegative self-adjoint operator H. Verify that

$$T_t T_s = T_{t+s}, \qquad \left.\frac{d}{dt}\right|_{t=0} T_t v = -Hv$$

for v in the domain of H. Conversely, state the hypotheses under which a strongly continuous self-adjoint contraction semigroup has the form $T_t = e^{-tH}$ for a unique nonnegative self-adjoint H. Show that if T_t is trace class for every $t > 0$, then H has compact resolvent [Dav89].

We add supersymmetry as the final layer of structure on a 1-dimensional field theory. To spell this out in detail would require a detour into the world of supermanifolds [DM99]. We instead highlight the key structures used in a way that makes contact with the previous lecture, referring to [HST10, ST11] for details. Supersymmetry upgrades the semigroup of intervals (7) to a *super semigroup*,

$$\begin{array}{c} + \\ \Big| {\scriptstyle (t,\,\theta)} \\ + \end{array} \longmapsto e^{-tH+\theta Q}, \qquad (t,\theta) \in \mathbb{R}^{1|1}_{\geq 0} \tag{8}$$

where $\mathbb{R}^{1|1}_{\geq 0} \subset \mathbb{R}^{1|1}$ is a subsemigroup of the super Lie group with multiplication

$$(t,\theta)\cdot(t',\theta') = (t + t' + \theta\theta', \theta + \theta'), \qquad (t,\theta),(t',\theta') \in \mathbb{R}^{1|1}.$$

Above, t is an even coordinate on $\mathbb{R}^{1|1}$ and θ is an odd coordinate. Under the same analytic conditions that guarantee the existence of a Hamiltonian, a representation of the super semigroup $\mathbb{R}^{1|1}_{\geq 0}$ is determined by an odd operator Q with $Q^2 = H$ generating the representation of $\mathbb{R}_{\geq 0} \subset \mathbb{R}^{1|1}_{\geq 0}$. To indicate the mechanism behind this, the super Lie algebra of this super Lie group is $1|1$-dimensional as a super vector space with basis $\partial_t, \partial_\theta - \theta\partial_t$ and brackets

$$[\partial_t, \partial_t] = [\partial_t, \partial_\theta - \theta\partial_t] = 0\,, \qquad [\partial_\theta - \theta\partial_t, \partial_\theta - \theta\partial_t] = -2\partial_t\,.$$

Hence, at the level of Lie algebras a representation of (8) is indeed determined by a single odd operator, denoted Q. The geometry of super intervals is essentially the only new ingredient brought by supersymmetry. One important consequence is integrality of the partition function [MS67]

$$\mathrm{sTr}(e^{-tH+\theta Q}) = \mathrm{Ind}(Q) \in \mathbb{Z}.$$

In particular, the Ramond sector partition function is constant on the moduli space of (super) circles.

Example 2.14 (Supersymmetric quantum mechanics)**.** Let M be a closed Riemannian spin manifold. Define a reflection positive $1|1$-dimensional supersymmetric field theory by

$$V = \Gamma(M;\mathbb{S}), \quad Q = \not{D}, \quad \langle f, g\rangle = \int (\overline{f}, g)\, d\mathrm{vol}$$

where $\mathbb{S} \to M$ is the spinor bundle, $\not{D}$ is the Dirac operator, and $(-,-)$ is the inner product on spinors. This determines a self-adjoint supersemigroup representation (8) on V that after Hilbert completion recovers supersymmetric quantum mechanics on M in the sense of Example 1.42.

Exercise 2.15 (The $1|1$ super-semigroup)**.** Carefully tracking the Koszul signs, verify that

$$e^{-tQ^2+\theta Q}e^{-t'Q^2+\theta' Q} = e^{-(t+t'+\theta\theta')Q^2+(\theta+\theta')Q}.$$

Conclude that $Q^2 = H$ is precisely the relation required for this to define a representation of the stated $1|1$ super-semigroup. *Hint:* Taylor expand in the θ coordinate and use $\theta^2 = 0$.

Problem 5 ($\star$)**.** Show that representations of the $1|1$-dimensional bordism category give a geometric model for K-theory. The arguments in [HST10] are mostly complete, but there are some small technical gaps. One notable deviation from the presentation above is that [HST10] use unoriented bordisms, real KO-theory, and subsequently a nonstandard version of positivity in place of the reflection positive theories sketched above.

Problem 6 ($\star\star$)**.** Construct the spin orientation in this framework: show that a (geometric) family of spin manifolds $M \to X$ determines an X-family of $1|1$-dimensional field theories, expanding the sketch from [ST04, Remark 3.2.21].

2.4. **2-dimensional theories and semigroups of annuli.** Next we indicate how the above structures in 1-dimensional quantum mechanical theories generalize in the setting of 2-dimensional quantum field theory. The classification of (oriented) 2-dimensional topological field theories is well-known, leading to commutative Frobenius algebras [Koc03]. Below we will focus exclusively on the non-topological case, and specifically the passage from the semigroups of intervals in the 1-dimensional theory to semigroups of *annuli* in the 2-dimensional case.

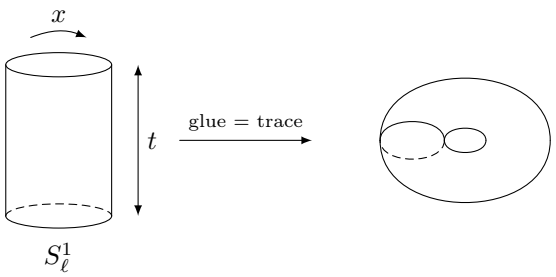


FIGURE 7. The annulus $A_{\ell,t,x}$ and the torus obtained by gluing its boundary circles. The small inscribed circle is the geodesic gluing locus.

Specifically, consider the category whose morphisms are annuli with a flat metric and parametrized geodesic boundary. The moduli space of such is 3-dimensional: after fixing the circumference $\ell \in \mathbb{R}_{>0}$ of the incoming and outgoing circle $S^1_\ell = \mathbb{R}/\ell\mathbb{Z}$, the annulus has a height $t \in \mathbb{R}_{\geq 0}$ and the difference in parametrization of the incoming and outgoing circles given by $x \in \mathbb{R}/\ell\mathbb{Z}$. Composition is

$$A_{\ell,t,x} \circ A_{\ell,t',x'} = A_{\ell,t+t',x+x'}$$

and hence for each $\ell \in \mathbb{R}_{>0}$, we obtain a semigroup of annuli $\mathbb{H}/\ell\mathbb{Z}$ for $\mathbb{H} \subset \mathbb{R}^2 \simeq \mathbb{C}$ the (closed) upper half plane.

A 2-dimensional Riemannian field theory restricts to such annuli

Annuli	$\hookrightarrow$ $\mathrm{Bord}_2(\mathrm{Riem})$ $\xrightarrow{Z}$	Vect
$S^1_\ell = \mathbb{R}/\ell\mathbb{Z}$	$\longmapsto$	V_ℓ
$A_{\ell,t,x}$	$\longmapsto$	$e^{-tH+ixP}\colon V_\ell \to V_\ell$

where V_ℓ denotes the vector space carrying the representation of the semigroup $\mathbb{H}/\ell\mathbb{Z}$ associated with the circle S^1_ℓ of circumference $\ell \in \mathbb{R}_{>0}$. Reflection positivity both provides a metric on V_ℓ and a condition on the semigroup representation: the (spatial) circle $\mathbb{R}/\ell\mathbb{Z} \subset \mathbb{H}/\ell\mathbb{Z}$ acts unitarily, while the (temporal) positive imaginary axis $i\mathbb{R}_{\geq 0} \subset \mathbb{H} \subset \mathbb{C}$ acts by self-adjoint operators.[2] In good analytical situations, we get self-adjoint infinitesimal generators $P(\ell)$ and $H(\ell)$ for these respective actions and

$$Z(A_{\ell,t,x}) = e^{ixP(\ell)-tH(\ell)} \tag{9}$$

where $P(\ell)$ is the *momentum operator* and $H(\ell)$ is the Hamiltonian, as before. The spectrum of $P(\ell)$ must be contained in $\frac{2\pi}{\ell}\mathbb{Z}$ since $P(\ell)$ generates an S^1_ℓ-action.

Fact 2.16. *The value of Z on the torus is the trace of the value on the annulus as in Figure 7, i.e.,*

$$t > 0 \implies Z(A_{\ell,t,x}) \text{ is trace class.} \tag{10}$$

Remark 2.17. The precise meaning of "trace class" depends on the analytical target, e.g., in the framework of [ST11], one finds that the value $Z(A_{\ell,t,x})$ is a nuclear operator.

[2]The asymmetry comes from the fact that reflection positivity can be seen as the composition of complex conjugation on $\mathbb{H}$ followed by a rotation by π; this leaves the imaginary axis fixed while reversing the real axis, resulting in the operator e^{-tH} being equal to its adjoint, while the operator e^{-iP} is the inverse of its adjoint.

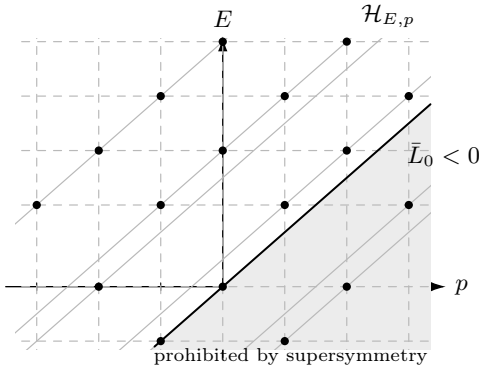


FIGURE 8. Visualizing a representation of the 2-dimensional semigroup of annuli: the commuting operators H and P have finite-rank eigenspaces $\mathcal{H}_{E,p}$ with energy E the eigenvalue of H and momentum p the eigenvalue of P, which we indicate by a dot with coordinate (p, E). The dashed lines represent constant values of p or E, which are equally spaced for the momenta p (by Exercise 2.19) but need not have any specific spacing for the energies E. Since the eigenspaces of H are finite-dimensional, the total contribution from each horizontal line is a finite-dimensional subspace of $\mathcal{H}$. The diagonal lines correspond to eigenvalues of $\overline{L}_0(\ell)$, with eigenspaces the direct sum of the dots on the corresponding line. Adding supersymmetry provides an odd square root of $\overline{L}_0(\ell)$. This imposes conditions on the spectrum: $\overline{L}_0(\ell)$ cannot have negative eigenvalues, which prohibits nonzero labels in the shaded region below the $\bar{L}_0 = 0$ line.

There are different ways to see the above fact, but a particularly robust perspective comes from the theory of categorical traces developed in [ST12]. The trace constructed there comes from generalizing dualizability arguments in topological field theories to the geometric (non-dualizable) setting. A related modern account is the theory of "rigid firm" categories of Theo Johnson-Freyd's lecture course in this series [Ber26].

The condition (10) has many important consequences for 2-dimensional quantum field theory. For the time being, we apply the observation from quantum mechanics: $H(\ell)$ has spectrum bounded below and its eigenspaces are finite-dimensional, see Exercise 1.8. Using that the (unbounded) operators $H(\ell)$ and $P(\ell)$ (strongly) commute, they are simultaneously diagonalizable. This leads to a configuration space description of (9) where a point $(p, E) \in \mathbb{R}^2$ is labeled by the subspace $\mathcal{H}_{p,E} \subset \mathcal{H}$ that is the intersection of the p-eigenspace of P and the E-eigenspace of H; see Figure 8. The fact that H has compact resolvent means that the direct sum of eigenspaces at a fixed horizontal line of height E results in a finite-dimensional subspace of $\mathcal{H}$.

For categories of annuli with spin structures, the two spin structures on the circle lead to a pair of semigroup representations:

Spin Annuli	$\hookrightarrow$ $\mathrm{Bord}_2^{\mathrm{Spin}}(\mathrm{Riem})$ $\xrightarrow{Z}$	sVect
$S^1_{\ell,p}$ $S^1_{\ell,ap}$	$\longmapsto$	$V_{\ell,p}$ $V_{\ell,ap}$
$A_{\ell,p,t,x}$ $A_{\ell,ap,t,x}$	$\longmapsto$	$Z(A_{\ell,p,t,x})\colon V_{\ell,p} \to V_{\ell,p}$ $Z(A_{\ell,ap,t,x})\colon V_{\ell,ap} \to V_{\ell,ap}$

These representations have similar descriptions to (9) with two modifications: (i) the vector spaces $V_{\ell,p}$ and $V_{\ell,ap}$ are $\mathbb{Z}/2$-graded, and (ii) the action of the momentum operator is slightly different as described in Exercise 2.19. There are four numbers we can extract from these annuli,

$$\mathrm{sTr}(Z(A_{\ell,p,t,x})), \quad \mathrm{Tr}(Z(A_{\ell,p,t,x})), \quad \mathrm{Tr}(Z(A_{\ell,ap,t,x})), \quad \mathrm{sTr}(Z(A_{\ell,ap,t,x})), \tag{11}$$

identified with the different tori one obtains by gluing annuli along their boundary as in Figure 7. The four values correspond to the four isomorphism classes of spin structure on the torus, e.g., via $H^1(T^2; \mathbb{Z}/2) = \mathbb{Z}/2 \oplus \mathbb{Z}/2$ [Ati71]. The action by isometries of tori must leave the values (11)

invariant. There are two orbits for this action, corresponding to the odd spin structure (and $\mathrm{sTr}(Z(A_{\ell,p,t,x}))$) and the even spin structure (which permutes the remaining 3 values). Below we will be primarily interested in the odd spin structure, which in turn draws our focus to the periodic (or Ramond) sector in the representation of annuli. To simplify notation, we write

$$V_\ell := V_{\ell,p}, \qquad A_{\ell,t,x} := A_{\ell,p,t,x}, \qquad Z(T_{\ell,t,x}) := \mathrm{sTr}(Z(A_{\ell,p,t,x}))$$

throughout.

Remark 2.18. There is a larger semigroup of annuli with a conformal structure, whose representations are closely related to positive-energy projective representations of $\mathrm{Diff}^+(S^1)$. The Lie algebra of this group is essentially the Virasoro algebra [Seg04].

Exercise 2.19 (Spin structures and momentum)**.** Suppose that $P(\ell)$ generates rotations of a circle of circumference ℓ. Show that $e^{i\ell P(\ell)} = 1$ implies

$$\mathrm{spec}(P(\ell)) \subset \frac{2\pi}{\ell}\mathbb{Z}.$$

If instead $e^{i\ell P(\ell)} = (-1)^F$, show that the even states have momentum in $\frac{2\pi}{\ell}\mathbb{Z}$ and the odd states have momentum in $\frac{2\pi}{\ell}(\mathbb{Z} + \frac{1}{2})$. The case $e^{i\ell P(\ell)} = 1$ corresponds to the periodic (or Ramond) circle, while $e^{i\ell P(\ell)} = (-1)^F$ is the antiperiodic (or Neveu–Schwarz) circle.

2.5. **Representations of annuli in 2-dimensional theories with $\mathcal{N} = (0,1)$ supersymmetry.** Supersymmetry can again be described in terms of supergeometry, where one takes the super semigroup $\mathbb{H}^{2|1} \subset \mathbb{R}^{2|1}$ where $\mathbb{R}^{2|1}$ is the super Lie group with multiplication

$$(\tau, \bar\tau, \theta) \cdot (\tau', \bar\tau', \theta') = (\tau + \tau', \bar\tau + \bar\tau' + \theta\theta', \theta + \theta'), \quad (\tau, \bar\tau, \theta), (\tau', \bar\tau', \theta') \in \mathbb{R}^{2|1} \tag{12}$$

where τ and $\bar\tau$ restrict to complex coordinates on the reduced plane $\mathbb{R}^2 \subset \mathbb{R}^{2|1}$, and θ is an odd coordinate. The super Lie algebra of this super Lie group is spanned by even elements $\partial_\tau, \partial_{\bar\tau}$ and an odd element $\partial_\theta - \theta\partial_{\bar\tau}$. The (super) Lie brackets are

$$[\partial_\theta - \theta\partial_{\bar\tau}, \partial_\tau] = [\partial_\theta - \theta\partial_{\bar\tau}, \partial_{\bar\tau}] = [\partial_\tau, \partial_{\bar\tau}] = 0, \quad [\partial_\theta - \theta\partial_{\bar\tau}, \partial_\theta - \theta\partial_{\bar\tau}] = 2(\partial_\theta - \theta\partial_{\bar\tau})^2 = -2\partial_{\bar\tau}$$

and hence the super Lie algebra of (12) is generated by the pair $\partial_\theta - \theta\partial_{\bar\tau}$ and ∂_τ. In particular, a representation of (12) is determined by even operators L_0 and $\bar L_0$ as well as an odd operator $\bar G_0$ satisfying the relations

$$[\bar G_0, L_0] = [\bar G_0, \bar L_0] = [L_0, \bar L_0] = 0\,, \qquad [\bar G_0, \bar G_0] = 2\bar L_0. \tag{13}$$

As in our treatment of quantum mechanics, rather than dive into the supergeometry we will unpack the algebraic consequences for representations of annuli. This will come down to super Lie algebra representations (13) satisfying additional properties. To get started, define

$$\tau := \frac{x + it}{\ell} \in \mathbb{H}/\mathbb{Z}, \qquad q = e^{2\pi i\tau} \tag{14}$$

which repackages (9) in terms of this complex coordinate

$$Z(A_{\ell,t,x}) = q^{L_0(\ell)}\overline{q}^{\overline{L}_0(\ell)}, \qquad L_0(\ell) = \frac{\ell}{4\pi}(H(\ell) + P(\ell)), \quad \overline{L}_0(\ell) = \frac{\ell}{4\pi}(H(\ell) - P(\ell)).$$

for self-adjoint operators $L_0(\ell)$ and $\bar L_0(\ell)$. From (13), supersymmetry requires an odd square root

$$\bar G_0(\ell)^2 = \bar L_0(\ell) \iff Q(\ell)^2 = \frac{1}{2}(H(\ell) - P(\ell)), \quad \bar G_0(\ell) = \sqrt{\frac{\ell}{2\pi}}Q(\ell)$$

or equivalently the odd operators $Q(\ell)$. This gives

$$Z(A_{\ell,t,x,\theta}) = e^{-tH + ixP + \sqrt{2\pi i}\,\theta\bar G_0(\ell)} = q^{L_0(\ell)}\overline{q}^{\bar L_0(\ell)}e^{\sqrt{2\pi i}\,\theta\bar G_0(\ell)} \colon V_\ell \to V_\ell \tag{15}$$

where the gluing law for super annuli is inherited from the multiplication (12). Again, (15) is trace-class and $H(\ell), P(\ell), Q(\ell)$ are self-adjoint operators. The configuration space pictures for these super annuli representations can always be restricted to $\theta = 0$ to recover a representation of the usual annuli as in Figure 8. The data of a supersymmetry forces $\bar L_0(\ell)$ to be nonnegative, prohibiting nontrivial eigenspaces below the line $E = p$, i.e., where $\bar L_0 < 0$. Furthermore, on a line where $E - p$ is constant and nonzero, the supersymmetry gives an isomorphism between the even and odd parts of this eigenspace, see Exercise 2.22.

Remark 2.20. We emphasize that this data is just for the Ramond sector (odd spin structure). The Neveu-Schwarz sector (even spin structure) does not have an odd operator Q acting on its space of states: the spinor bundle in this sector has monodromy -1 (see Exercise 2.19), which prevents the local supersymmetry operator Q from descending to the circle.

Remark 2.21. We are only considering $\mathcal{N} = (0,1)$ supersymmetry in these notes. The general case of $\mathcal{N} = (\mathcal{N}_L, \mathcal{N}_R)$ supersymmetry is the data of $\mathcal{N}_L$ (super) commuting odd square roots of $L_0(\ell)$ and $\mathcal{N}_R$ (super) commuting odd square roots of $\bar{L}_0(\ell)$.

Exercise 2.22 (The supersymmetric energy bound)**.** Let v be a simultaneous eigenvector of H and P with eigenvalues E and p, and suppose

$$Q^2 = \frac{1}{2}(H - P).$$

Show that $E \geq p$. If $E > p$, prove that Q gives an isomorphism between the even and odd parts of the joint (E,p)-eigenspace, and write down its inverse.

Problem 7 ($\star$)**.** Spell out the configuration description for the Neveu-Schwarz sector, and translate the conditions (11) into restrictions on pairs of configurations, one in the Neveu–Schwarz sector and one in the Ramond sector as in Figure 8.

3. Supersymmetric field theories and topological modular forms

In the prior lecture we saw that the value of a 2-dimensional field theory on a positive volume annulus gives a trace-class operator (10). This affords nice analytical properties for the theory, namely the energy eigenspaces of the Hamiltonian decompose the infinite-dimensional Hilbert space into a direct sum of finite-dimensional subspaces. This is a generic feature of quantum field theory on compact spaces: energy cutoffs take an a priori infinite-dimensional problem and filter it by finite-dimensional ones. Many problems can be solved by looking at a finite-dimensional (finite energy) chunk, and then understanding modifications coming from incorporating higher energy states. This is the jumping-off point for (Wilsonian) effective field theory, which is one of the most important concepts in quantum field theory as actually practiced by physicists [Pol84].

The trace property (10) has another important consequence. The value of the theory on the torus must be invariant under isometries, which leads to an $\mathrm{SL}_2(\mathbb{Z})$-invariance condition. Unpacking this condition provides the entry point for modular forms in the story.

3.1. The value of a 2-dimensional theory on a torus.

Definition 3.1. The (odd or Ramond–Ramond) *partition function* of a 2-dimensional supersymmetric field theory is the super trace of its value on annuli

$$Z(\ell, \tau, \bar{\tau}, \theta) := \mathrm{sTr}(e^{-tH(\ell)+ixP(\ell)+\theta Q(\ell)}) \in C^\infty((\mathbb{R}_{>0} \times \mathbb{H}^{2|1})/\mathbb{Z})$$

in the notation from (15).

Proposition 3.2. *The Ramond–Ramond partition function of a $\mathcal{N} = (0,1)$ supersymmetric theory depends holomorphically on $\mathbb{H}$.*

Proof. Taylor expanding the (trace class) semigroup in the odd variable θ,

$$\begin{aligned} Z(\ell, \tau, \bar{\tau}, \theta) &= \mathrm{sTr}\left(e^{-tH(\ell)+ixP(\ell)}(1 + \theta Q(\ell))\right) \\ &= \mathrm{sTr}\left(e^{-tH(\ell)+ixP(\ell)}\right) + \theta\,\mathrm{sTr}\left(Q(\ell)e^{-tH(\ell)+ixP(\ell)}\right) \\ &= \mathrm{sTr}\left(e^{-tH(\ell)+ixP(\ell)}\right), \end{aligned}$$

where in the last line we use that the super trace of an odd endomorphism is zero. Holomorphy then follows from changing coordinates (14) and taking a derivative,

$$\begin{aligned}\partial_{\overline{\tau}}\,\mathrm{sTr}(q^{L_0}\overline{q}^{\bar{L}_0}) &= -2\pi i\,\mathrm{sTr}(\bar{L}_0 q^{L_0}\overline{q}^{\bar{L}_0})\\ &= -2\pi i\,\mathrm{sTr}(\overline{G_0}^2 q^{L_0}\overline{q}^{\bar{L}_0})\\ &= -\pi i\,\mathrm{sTr}([\overline{G_0},\overline{G_0}q^{L_0}\overline{q}^{\bar{L}_0}])\\ &= 0\end{aligned}$$

using that the super trace of a super commutator vanishes, see Exercise 3.5. □

We recall that a *modular function* is a holomorphic function on the upper half plane $f \in \mathcal{O}(\mathbb{H})$ with the $\mathrm{SL}_2(\mathbb{Z})$-invariance property

$$f\left(\frac{a\tau+b}{c\tau+d}\right) = f(\tau), \qquad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z})$$

and a finite-order pole for $\tau \to i\infty$. A modular function is *integral* if it further has an integral q-expansion,

$$f(q) \in \mathbb{Z}(\!(q)\!) \subset \mathbb{C}(\!(q)\!), \qquad q = e^{2\pi i\tau}.$$

The finiteness of the pole at $\tau \to i\infty$ corresponds to the Laurent expansion of $f(q)$ having finitely many negative powers of q.

Theorem 3.3 ([ST11, Theorem 1.15 and §3-4])**.** *The Ramond–Ramond partition function of an $\mathcal{N} = (0,1)$ supersymmetric theory is an* integral *modular function.*

Proof. We observe that $L_0|_{\ker \bar{L}_0} = \frac{\ell}{2\pi}P$ has integral spectrum. Then we compute

$$\begin{aligned}\mathrm{sTr}(q^{L_0}\overline{q}^{\bar{L}_0}) &= \mathrm{sTr}(q^{L_0}|_{\ker(\bar{L}_0)})\\ &= \sum_k \mathrm{sTr}(q^{L_0}|_{\mathcal{H}_k}), \qquad \mathcal{H}_k := \ker(\bar{L}_0)\cap\ker(L_0-k)\\ &= \sum_{k\geq -N} q^k\,\mathrm{sdim}(\mathcal{H}_k) \in \mathbb{Z}(\!(q)\!)\end{aligned}$$

where in the first line, the existence of the odd square root $\bar{G}_0^2 = \overline{L}_0$ implies that the nonzero eigenspaces of $\bar{L}_0$ do not contribute to the super trace, see Exercise 3.6. This supplies the claimed integrality.

To see modularity, we claim

$$Z(\ell,\tau,\overline{\tau},\theta) = Z(\tau) = Z\left(\frac{a\tau+b}{c\tau+d}\right). \tag{16}$$

The first equality is Proposition 3.2 (for independence of θ and $\bar{\tau}$) and the q-expansion argument of the previous paragraph (for independence of ℓ). The second equality in (16) follows by the isometry of flat (Euclidean) tori,

$$T_{|c\tau+d|\ell,\,\frac{a\tau+b}{c\tau+d}} \cong T_{\ell,\tau} \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}), \tag{17}$$

where $T_{\ell,\tau} := \mathbb{C}/(\ell\mathbb{Z}\oplus\ell\tau\mathbb{Z})$ for $\ell \in \mathbb{R}_{>0}, \tau \in \mathbb{H}$. □

Exercise 3.4. Confirm that the tori (17) are isometric. *Hint:* The lattices defining the tori are related by a change of basis in $\mathrm{SL}_2(\mathbb{Z})$.

Exercise 3.5 (Supertraces and holomorphy)**.** Let A and B be homogeneous operators for which the relevant products are trace class. Prove that

$$\mathrm{sTr}([A,B]) = 0, \qquad [A,B] = AB - (-1)^{|A||B|}BA.$$

Exercise 3.6 (Supersymmetric cancellation, 2-dimensional version)**.** The following is a generalization of Exercise 1.45. Let V_λ be the λ-eigenspace of $\bar{L}_0 = \bar{G}_0^2$. Show that, when $\lambda > 0$,

$$\bar{G}_0 \colon V_\lambda^{\mathrm{even}} \longrightarrow V_\lambda^{\mathrm{odd}}$$

is an isomorphism. Deduce that $\mathrm{sdim}(V_\lambda) = 0$ and hence $\mathrm{sTr}(q^{L_0}\overline{q}^{\bar{L}_0}) = \mathrm{sTr}\big(q^{L_0}|_{\ker \bar{L}_0}\big)$.

3.2. **Anomalous and relative field theories.** Viewing a quantum field theory as a kind of representation, an anomalous quantum field theory should be thought of as a *projective* representation. To set up this analogy, a functor $\mathbb{B}G \to \mathrm{Vect}$ selects a vector space with G-action, i.e., a G-representation. A projective G-representation is equivalent to a natural transformation

$$\mathbb{B}G \;\underset{\alpha}{\overset{\mathbb{1}}{\Rightarrow}}\; \mathrm{Bimod} \qquad (Z \Downarrow) \tag{18}$$

where α is a $\otimes$-invertible functor valued in the (Morita) bicategory Bimod of algebras, bimodules, and bimodule maps.

Remark 3.7. The setup (18) and subsequent generalizations parse for any *categorical delooping* of Vect, meaning a symmetric monoidal 2-category $\mathcal{C}$ with $\mathrm{End}_{\mathcal{C}}(\mathbb{1}) \simeq \mathrm{Vect}$. The $\mathbb{Z}/2$-graded variant sBimod of the Morita bicategory is in some sense the "correct" target for *topological* field theories [FH21]. The subtle functional analysis inherent in nontopological quantum theories both presents many constraints and a zoo of choices for such deloopings. Finding good candidates is an active area of research. André Henriques's lecture course in this series treats (super) von-Neumann-algebras and their further delooping to *bicommutant categories* as targets for 2- and 3-extended theories.

Exercise 3.8. Show that the data in (18) is equivalent to a projective G-representation.

(1) Suppose that $\alpha(\mathrm{pt}) = \mathbb{C}$. Then $\alpha(g) = L_g$ is a $\otimes$-invertible $\mathbb{C}$-$\mathbb{C}$-bimodule. Show that the 2-functoriality of α determines a group cohomology cocycle $c \in C^2(G; \mathbb{C}^\times)$ which is well-defined up to a coboundary. *Hint:* Fix isomorphisms $\mathbb{C} \cong L_g$ so that the remaining data is $\mathbb{C}^\times$-valued.
(2) Let $V = Z(\mathrm{pt}) \in \mathrm{End}_{\mathrm{Bimod}}(\mathbb{C})$ be the vector space assigned to the unique object of $\mathbb{B}G$. Show that the values $Z(g)$ determine a projective action of G on V for the central extension constructed in (1).

Definition 3.9 ([ST11, §5])**.** An *anomalous quantum field theory with anomaly* α is a $\otimes$-invertible monoidal functor α and (op)lax monoidal natural transformation Z,

$$\mathrm{Bord}_d(\mathcal{F}) \;\underset{\alpha}{\overset{\mathbb{1}}{\Rightarrow}}\; \mathrm{Bimod} \qquad (Z \Downarrow) \tag{19}$$

We unpack Definition 3.9 as follows. The anomaly α assigns

$$\alpha(Y) \text{ a Morita invertible algebra}, \qquad {}_{\alpha(Y_{\mathrm{in}})}\alpha(\Sigma)_{\alpha(Y_{\mathrm{out}})} \text{ an invertible bimodule}$$

for each object $Y \in \mathrm{Bord}_d(\mathcal{F})$ and morphism $\Sigma\colon Y_{\mathrm{in}} \to Y_{\mathrm{out}}$. There is further data supplied by maps of bimodules associated with compositions of bordisms, analogous to the 2-cocycle in the motivating example (18). Roughly, these data correspond to a central extension of the bordism category.

The anomalous field theory Z assigns

$$Z(Y) \text{ a right } \alpha(Y)\text{-module}, \qquad Z(\Sigma)\colon Z(Y_{\mathrm{in}}) \otimes_{\alpha(Y_{\mathrm{in}})} \alpha(\Sigma) \to Z(Y_{\mathrm{out}}) \text{ a right } \alpha(Y_{\mathrm{out}})\text{-module map}$$

which gives the analog of a projective representation of $\mathrm{Bord}_d(\mathcal{F})$ for central extension specified by α.

Remark 3.10. We tighten the relationship between (18) and (19) in Example 4.4 below.

Remark 3.11. One defines reflection positivity for anomalous field theories in much the same way as before: enhance the C_2-action by reversal of complex structures to an action on Bimod, and ask for a C_2-equivariant functor α for the same C_2-action on bordisms. Unpacking this, a reflection structure for an anomaly α should be a $*$-structure on the algebras associated to each object, and the positivity condition should be suitable C^*-conditions.

Remark 3.12. Anomalies are often restrictions of once extended invertible $(d+1)$-dimensional theories

$$\mathrm{Bord}_{d,d-1}(\mathcal{F}) \hookrightarrow \mathrm{Bord}_{d+1,d,d-1}(\mathcal{F}) \xrightarrow{\mathcal{I}} \mathrm{Bimod} \tag{20}$$

where $\mathrm{Bord}_{d,d-1}(\mathcal{F}) = \mathrm{Bord}_d(\mathcal{F})$ is the bordism category as we have been considering above, while $\mathrm{Bord}_{d+1,d,d-1}(\mathcal{F})$ denotes a bordism 2-category with objects $(d-1)$-manifolds, 1-morphisms d-manifolds and 2-morphisms $(d+1)$-manifolds. The restriction of the once extended theory $\mathcal{I}$ to d-manifolds and $(d-1)$-manifolds gives a once categorified d-dimensional theory. When $\mathcal{I}$ is invertible, this restriction defines an anomaly. This is the setup for *anomaly inflow* [Fre14].

Remark 3.13. Stolz and Teichner refer to anomalous theories as *twisted field theories*, owing to their relevance to twisted cohomology under Conjecture A (as we will discuss below). This presents a terminological collision for the usage of topological twistings in lectures presented at this summer school (in mirror symmetry and elsewhere). As such we have opted for the terminology above.

Problem 8 $(\star)$**.** Spell out reflection positivity data and the positivity condition for anomalies and anomalous theories in the sense of Definition 3.9. Beware of the sign differences when comparing the C^*-algebra literature with super algebras, and consider using the formalism of *stellar algebras* developed in [MS25, §5.2].

3.3. **K-theory and $1|1$-dimensional anomalous theories.** The nth (complex) Clifford algebra Cl_n is a Morita-invertible super algebra with $\otimes$-inverse Cl_{-n}. Assigning Cl_n to the value on a point determines a fully-extended 2-dimensional spin topological field theory that we can restrict to 1-dimensional spin manifolds

$$\mathrm{Bord}_1^{\mathrm{Spin}} = \mathrm{Bord}_{1,0}^{\mathrm{Spin}} \hookrightarrow \mathrm{Bord}_{2,1,0}^{\mathrm{Spin}} \xrightarrow{(\mathrm{Arf})^{\otimes n}} \mathrm{sBimod} \tag{21}$$

following (20), where the second arrow uses that dualizability (implied by invertibility) determines the value of the theory on all other bordisms. [3] These values on points and 1-manifolds are:

pt_+	$\longmapsto$	Cl_n
pt_-	$\longmapsto$	$\mathrm{Cl}_n^{\mathrm{op}}$
	$\longmapsto$	$\mathrm{Cl}_n{}_{\mathrm{Cl}_n \otimes \mathrm{Cl}_n^{\mathrm{op}}}$
	$\longmapsto$	${}_{\mathrm{Cl}_n}\mathrm{Cl}_{n\,\mathrm{Cl}_n}$
	$\longmapsto$	$\mathrm{HH}_0(\mathrm{Cl}_n, \mathrm{Cl}_n)$.

The values on intervals repackage Cl_n as a bimodule over itself, and the value on circles is the 0th Hochschild homology in the graded sense, see Exercise 3.15. The action of Cl_n on itself is the evident one, or twisted by $(-1)^F$, according to the spin structure on the circle. Endowing (21) with a reflection structure uses the data of a $*$-structure $*\colon \overline{\mathrm{Cl}}_n \cong \mathrm{Cl}_n^{\mathrm{op}}$. We promote (21) to a supersymmetric theory by declaring the value on a super interval of length (t, θ) to be independent of the geometric parameter and equal to Cl_n as a bimodule over itself.

Remark 3.14. When $n = 1$, the 2-dimensional theory in (21) is also called the *Arf theory* whose value on a closed Spin manifold is the $\{\pm 1\}$-valued Arf invariant. A very nice account of this theory is given in [Gun16].

Exercise 3.15. Verify that every $A \in \mathrm{sBimod}$ is dual to A^{op}, with dualizing data gotten by regarding A as a bimodule over itself. By the cobordism hypothesis, there is a topological theory

$$Z_A\colon \mathrm{Bord}_{1,0}^{\mathrm{Spin}} \to \mathrm{sBimod}, \qquad Z_A(\mathrm{pt}_+) = A. \tag{22}$$

Compute the value of Z_A on the two spin circles. In the case $A = \mathrm{Cl}(V)$, show $Z_{\mathrm{Cl}(V)}(\bigcirc^p) \cong \Lambda^{\mathrm{top}} V$, the line that is trivialized by a volume element $\omega = e_1 \dots e_{\dim V}$, with parity equal to $\dim(V) \bmod 2$.

[3] Strictly speaking this only determines the framed theory. One needs to add Spin(2)-fixed point data.

Hint: Decomposing the circle as cup glued to a cap, we have

$$Z_A(\bigcirc^p) \cong A \otimes_{A\otimes A^{\mathrm{op}}} A^{\mathrm{op}} \cong A/[A,A] =: \mathrm{HH}_0(A,A)\,.$$

where we emphasize that the tensor product is the *graded* tensor product, the abelianization is the graded one, and $\mathrm{HH}_0(A,A)$ is similarly the 0th graded Hochschild homology group. The calculation for the anti-periodic circle is similar, with a $(-1)^F$ inserted.

Problem 9 ($\star$)**.** The restriction (21) does not immediately yield a supersymmetric anomaly theory. Construct a supersymmetric extension

$$\mathrm{Bord}_{1|1} \hookrightarrow \mathrm{Bord}_{2|(1,1),1|1,0|1} \xrightarrow{(\text{super Arf})^{\otimes n}} \mathrm{sBimod} \tag{23}$$

that makes this evident and that lifts the Arf theory to a fully-extended theory with $\mathcal{N} = (1,1)$ supersymmetry. The fact that 2-dimensional theories with $\mathcal{N} = (1,1)$ supersymmetry have 1-dimensional boundaries with $\mathcal{N} = 1$ supersymmetry was discussed in Davide Gaiotto's lecture in this series [Ber26].

Remark 3.16. The bordism category $\mathrm{Bord}_{1|1}$ has a (weak) involution that in terms of the supergeometry of $\mathbb{R}^{1|1}$ is determined by $(t,\theta) \mapsto (-t, i\theta)$. This automorphism squares to $(t,\theta) \mapsto (t,-\theta)$, the spin flip isomorphism in $\mathrm{Bord}_{1|1}$. In short, this geometry determines a functor

$$\mathrm{or}\colon \mathrm{Bord}_{1|1} \to \mathrm{Bord}_{1|1} \tag{24}$$

with $\mathrm{or}\circ\mathrm{or} \simeq \mathrm{id}$, i.e., a (weak) C_2-action on $\mathrm{Bord}_{1|1}$. This is essentially the orientation reversing automorphism on $\mathrm{Bord}_{1|1}$ (modulo a subtle aspect about real structures on supermanifolds). For any functor out of $\mathrm{Bord}_{1|1}$, we can ask for homotopy fixed point data relative to this C_2-action. In the case of the functor (21), such fixed point data is exactly a Real structure $\mathrm{Cl}_n \simeq \overline{\mathrm{Cl}}_n$ on the nth Clifford algebra.

Having fixed the anomaly determined by Cl_n, an anomalous theory Z

$$\mathrm{Bord}_{1|1} \;\underset{\mathrm{Cl}_n}{\overset{\mathbb{1}}{\overset{\Downarrow Z}{\rightrightarrows}}}\; \mathrm{sBimod} \tag{25}$$

assigns values

pt_+	$\longmapsto$	$\mathcal{H}_{\mathrm{Cl}_n}$
pt_-	$\longmapsto$	$\overline{\mathcal{H}}_{\overline{\mathrm{Cl}_n}}$
$\overset{\cap}{\pm\ \mp}$	$\longmapsto$	$(\overline{\mathcal{H}} \otimes_{\mathrm{Cl}_n^{\mathrm{op}}} \mathcal{H} \to \mathbb{C})$
$\begin{matrix}+\\ \vert\,{\scriptstyle (t,\theta)}\\ +\end{matrix}$	$\longmapsto$	$(e^{-tH+\theta Q} : \mathcal{H} \to \mathcal{H})$
$\bigcirc$	$\longmapsto$	$\mathrm{sTr}_{\mathrm{Cl}_n}(e^{-tH+\theta Q}) \in \mathrm{HH}_0(\mathrm{Cl}_n)^\vee$

where $\mathrm{HH}_0(\mathrm{Cl}_n)^\vee$ denotes the dual line.

Theorem 3.17 (Stolz–Teichner)**.** *The collection of* Cl_n*-anomalous theories gives geometric cocycles for* KU^n*. The involution* (24) *acts on the collection of such cocycles, giving the* C_2*-spaces in the* KR*-spectrum.*

Sketch. As indicated above, a Cl_n-anomalous theory determines a Clifford module $\mathcal{H}$ and an odd Clifford linear operator Q with compact resolvent.[4] The collection of these operators gives the nth space in the KU-spectrum. We identify the C_2-action induced by (24) with complex conjugation, hence recovering the spaces in the KR-spectrum. □

The sequence of spaces above does not yet recover K-theory: we need the connecting maps to get the full spectrum. As usual in a multiplicative cohomology theory, these connecting maps are determined by *suspension classes*, representing the generator in $\mathrm{KU}^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}) \simeq \mathbb{Z}$.

[4]Compact resolvent is basically the Fredholm condition for unbounded operators.

Example 3.18 (Suspension classes)**.** Let V be an inner product space. For each $v \in V$, define a $\mathrm{Cl}(V)$-anomalous theory as in (25) via

$$\mathcal{H} = \mathrm{Cl}(V), \quad Q_v = v \cdot - : \mathcal{H} \to \mathcal{H}$$

i.e., the supersymmetry generator Q is left Clifford multiplication by v, which commutes with the right $\mathrm{Cl}(V)$-action on $\mathrm{Cl}(V)$. Letting $v \in V$ vary, we obtain a V-family of operators Q_v determining a V-family of field theories. This family represents the suspension class

$$[\mathrm{Cl}(V), Q_v] \in \mathrm{KU}^V(V, V \setminus \{0\}). \tag{26}$$

To see that we obtain a relative class, note that Clifford multiplication by $v \neq 0$ is invertible, and therefore represents the zero class in K-theory. Taking $V = \mathbb{R}^n$ and setting $\mathrm{KU}^{\mathbb{R}^n} = \mathrm{KU}^n$ leads to the usual integer grading.

This is a continuation of Problems 5 and 8.

Problem 10 ($\star\star$)**.** Give a complete proof of Theorem 3.17, extending arguments in [HST10] to include reflection positive Cl_n-anomalous field theories.

3.4. **Sketch of the** $2|1$**-dimensional generalization.** The Clifford algebra $\mathrm{Cl}(V)$ arises from quantizing a free particle moving in the odd vector space ΠV [Fre99, pg 19]. This theory has higher dimensional generalizations called *free fermions*, which come from quantizing the space of maps from spin d-manifolds to ΠV. We caution that there are several distinct notions of these free fermions, even after specializing to a fixed dimension. Below we describe one version in dimension 2, following [ST11, §5-6].

There is a 2-dimensional Spin Riemannian field theory that takes values

$\mathrm{Bord}_2^{\mathrm{Spin}}$	$\xrightarrow{\mathrm{Fer}_V}$	sBimod
S^1 + spinor bundle $\mathbb{S} \to S^1$	$\longmapsto$	$\mathrm{Cl}(\Gamma(S^1, \mathbb{S} \otimes V))$
Σ spin manifold with Dirac operator $\not{D}$	$\longmapsto$	$\begin{cases} \mathrm{Fock}(\not{D}_V) & \Sigma \text{ connected, } \partial\Sigma \neq \emptyset \\ \mathrm{Pf}(\not{D}_V) & \Sigma \text{ closed.} \end{cases}$

To spell this out, the value of Fer_V on the circle is the Clifford algebra of sections of the twisted spinor bundle $\mathbb{S} \otimes V$, identifying V with the trivial bundle with fiber V on S^1. The value on a closed surface Σ is $\mathrm{Pf}(\not{D}_V)$, the fiber of the Pfaffian line of the V-twisted Dirac operator on Σ [Fre87, §3]. The value on a surface with boundary is the Fock space $\mathrm{Fock}(\not{D}_V)$ gotten by viewing the harmonic spinors on Σ as defining a Lagrangian (or Calderón) subspace of the boundary spinors

$$\mathrm{Fer}_V(\Sigma) := \mathrm{Fock}\left(\ker(\not{D}_V)|_{\partial\Sigma} \underset{\text{Lagrangian}}{\subset} \Gamma(\partial\Sigma, \mathbb{S} \otimes V)\right).$$

Creation and annihilation operators endow the completed Fermionic Fock space (roughly, exterior vector space) on this subspace with the structure of an irreducible module over the Clifford algebra of the boundary spinors, e.g., see [LR20] for details. Anomaly theories can always be tensored; for the anomaly theories above, we have isomorphisms

$$\mathrm{Fer}_V \otimes \mathrm{Fer}_W \cong \mathrm{Fer}_{V \oplus W}.$$

The close connection between the 2-dimensional (chiral) Dirac operator and the $\bar{\partial}$-operator reveals that this theory is a *chiral CFT*: its value on (Ramond) annuli gives a semigroup representation with $\bar{L}_0 = 0$. At least morally, this gives a canonical extension to an $\mathcal{N} = (0, 1)$ supersymmetric theory with $\bar{G}_0 = 0$.

For $V = \mathbb{R}$, the Pfaffian line $\mathrm{Pf}(\not{D})$ is 24-periodic when viewed as a line bundle over the moduli of (closed) genus 1 spin surfaces, meaning the 24th power of the Pfaffian line with $V = \mathbb{R}$ has a trivialization. Over higher genus surfaces, there is no periodicity: all powers are nontrivial. In view of Conjecture A, this suggests one restrict to *flat* surfaces to accommodate a periodicity. This sort of flat supersymmetric version of the fermions is what Stolz and Teichner set up in [ST11], giving a functor

$$\mathrm{Fer}_V \colon \mathrm{EBord}_{2|1} \longrightarrow \mathrm{sBimod}, \qquad \mathrm{EBord}_{2|1} := \mathrm{Bord}_{2|1}\begin{pmatrix} \text{flat} \\ \text{Riem} \end{pmatrix} \tag{27}$$

We write $\mathrm{Fer}_{\mathbb{R}^n} = \mathrm{Fer}_n$ below in the case that $V = \mathbb{R}^n$.

Theorem 3.19 (Stolz-Teichner)**.** *The functor* (27) *is 48-periodic,*

$$\mathrm{Fer}_k \simeq \mathrm{Fer}_{k+48}$$

whose periodicity generator has partition function Δ^2*, the square of the modular discriminant.*

[ST11, Theorem 1.16] states periodicity for 2|0-dimensional theories. The claimed extension to 2|1 anomaly functors requires justification.

Problem 11 ($\star\star\star$)**.** Construct a fully extended version of (27) that is $24^2 = 576$ periodic, with periodicity generator a field theory with partition function Δ^{24}.

To set up the above problem precisely, one needs a fully-extended (2-categorical) version of $\mathrm{Bord}_{2|1}$ and a 3-categorical enhancement of sBimod. Then one needs to construct values of a functor on points, intervals, and 2-manifolds with corners that recover (27) when restricted to closed 1-manifolds and 2-manifolds with boundary. Generalizing Theorem 3.17 in this fully-extended context is expected to recover the following version of Conjecture A.

Conjecture B ([ST11, Conjecture 1.17])**.** The collection of fully-extended Fer_n-anomalous 2|1-dimensional flat Riemannian field theories

$$\mathrm{EBord}_{2|1,1|1,0|1} \xrightarrow[\mathrm{Fer}_n]{\mathbb{1} \quad Z \Downarrow} \mathcal{B} \tag{28}$$

can be assembled into a space that is homotopy equivalent to the nth space in the TMF spectrum. Above, $\mathcal{B}$ is a symmetric monoidal 3-category with $\Omega\mathcal{B} = \mathrm{sBimod}$.

Remark 3.20. The cited conjecture proposed a specific mechanism for constructing a *space* of field theories. First one enhances the bordism category as $\mathrm{EBord}_{2|1}(M)$, wherein bordisms are equipped with a map to a smooth manifold M, i.e., adds background scalar fields valued in M. For fully-extended d-dimensional field theories, the functor that assigns to M the groupoid of (anomalous) field theories of M is expected to be a d-stack. In this setting, field theories over M are *concordant* if there is a field theory over $M \times \mathbb{R}$ that restricts at $\{0\}$ and $\{1\}$ to the given theories. Concordance classes of field theories are then representable by a space [GP20], and [ST11, Conjecture 1.17] asserts that this space represents TMF^n. There is one additional technical modification in this approach: the target category is replaced by a certain category of sheaves, see [ST11, Remark 3.16] and [Ulr21].

The spectrum structure on the spaces above is expected to come from a generalization of (26).

Conjecture C. Taking $\mathrm{Fer}(V)$ as a bimodule over itself, there is a V-family of supersymmetric deformations representing the suspension class in TMF,

$$[\mathrm{Fer}(V), Q_v] \in \mathrm{TMF}^V(V, V \setminus 0). \tag{29}$$

Example 3.21 (The Dirac operator on loop space)**.** The S^1-action on the loop space LM of a closed Riemannian d-manifold decomposes the normal bundle to the constant loops $M \subset LM$ into an infinite direct sum of finite-dimensional vector bundles on M [Wit88]. From the geometry of this normal bundle, Witten (formally) defines a localized spinor bundle with L^2-sections $\mathcal{H}_{LM}$ that is equipped with a Dirac operator $\not{D}_{LM}$ whose (formal) S^1-equivariant index is [Wit88, Wit87]

$$Z_M(\tau) = \mathrm{sTr}_{\mathcal{H}_{LM}}\left(q^{L_0}\overline{q}^{\overline{L}_0}\right) = \mathrm{Ind}_{S^1}(\not{D}_{LM}) = \eta(\tau)^{-d}\varphi_W(M),$$

where $\varphi_W(M;\tau)$ is the Witten genus,

$$\varphi_W(M) = \int_M \widehat{A}(TM)\ \mathrm{ch}\left(\bigotimes_{n\geq 1} S_{q^n}\left(T_{\mathbb{C}}M - \mathbb{C}^d\right)\right).$$

When M is a string manifold, Zagier [Zag86] confirmed Witten's expectation that $\eta(\tau)^{-d}\varphi_W(M)$ is a modular form.

Conjecture D. The operator $\not{D}_{LM} = Q$ is the supersymmetric generator for a Fer_{-d}-anomalous 2|1-dimensional field theory called the *supersymmetric sigma model* with target M. The image of this sigma model under Conjecture B is the image

$$[M] \in \pi_d \mathrm{MString} \to \pi_d \mathrm{TMF}$$

determined by the string orientation of TMF.

4. Towards equivariant elliptic objects and geometric cocycles for TMF

Field theories have a native notion of equivariant refinement, gotten by equipping bordisms with a G-bundle with connection, see Definition 4.1. In this way, Conjecture A has a completely canonical equivariant refinement. In this lecture we explain how this is expected to compare with equivariant elliptic cohomology.

This enhancement is interesting for at least four reasons. First, the equivariant structures make contact with classical objects like θ-functions, loop group representations, and modular tensor categories. This adds substantial depth to the already rich structures in elliptic cocycles. Second, equivariant structures in dimension 2 are very rigid, e.g., the theory of loop group representations at a fixed level shares a lot of features with the representation theory of *finite* groups. When setting up a framework for equivariant elliptic cocycles, this rigidity provides stringent constraints, making it harder to guess the wrong definitions. This can be useful for understanding the non-equivariant case. Third, U(1)-equivariant elliptic cohomology is essentially the theory of (derived) elliptic curves. It appears that understanding the corresponding U(1)-equivariant structures in quantum field theories reveals the role of (elliptic) group laws that are the homotopical underpinnings. Fourth and finally, combining elliptic twistings with equivariance leads to Lurie's *2-equivariant* elliptic cohomology [Lur09b, §5.1]. This theory enjoyed a (conjectured) universal property [Lur09b, §5.5], which charts a path toward constructing a map from field theories to elliptic cohomology required in Conjecture A.

4.1. Field theories with background G-symmetry.

Let $\mathbb{B}^\nabla G$ denote the stack classifying G-bundles with connection.

Definition 4.1. A field theory with *(background) G-symmetry* is a functor

$$\mathrm{Bord}_d(\mathcal{F} \times \mathbb{B}^\nabla G) \xrightarrow{Z} \mathrm{Vect}\,.$$

A field theory with *anomalous (background) G-symmetry* is a natural transformation

$$\mathrm{Bord}_d(\mathcal{F} \times \mathbb{B}^\nabla G) \underset{\alpha}{\overset{\mathbb{1}}{\overset{\curvearrowright}{\underset{\curvearrowright}{\;Z \Downarrow\;}}}} \mathrm{Bimod}\ . \tag{30}$$

Remark 4.2. The adjective *background* refers to the fact that the bordisms are equipped with a G-bundle, viewed as a *background field*. This is the only notion of symmetry we will deal with below. As such, we will drop this modifier and simply refer to G-symmetric field theories hereafter.

Remark 4.3. Reflection positivity is implemented in the same way as before. This often has the effect of promoting various linear G-actions into unitary representations; see Example 4.4 below.

Given an (anomalous) field theory with G-symmetry, we can always forget by precomposing with the functor that selects the trivial bundle with trivial connection,

$$\mathrm{Bord}_d(\mathcal{F}) \xrightarrow{\text{trivial bundle}} \mathrm{Bord}_d(\mathcal{F} \times \mathbb{B}^\nabla G). \tag{31}$$

Conversely, (as in Example 2.4) it is often interesting to ask whether a given field theory Z has an extension to one with G-symmetry,

$$\begin{array}{ccc} \mathrm{Bord}_d(\mathcal{F}) & \xrightarrow{\quad Z \quad} & \mathrm{Vect} \\ {\scriptstyle \mathrm{triv}}\big\downarrow & \nearrow & \\ \mathrm{Bord}_d(\mathcal{F} \times \mathbb{B}^{\nabla} G) & & \end{array} \tag{32}$$

with a similar diagram for the anomalous case. In general, equipping an anomaly or an anomalous field theory with a G-symmetry is extra data which can be obstructed, as we see in the following example.

Example 4.4 (Projective representations from 1d TFTs)**.** For our treatment of 1-dimensional (anomalous) topological field theories with G-symmetry, we first observe the commuting square

$$\begin{array}{ccc} \mathrm{pt} & \xrightarrow{\quad \mathrm{pt}_+ \quad} & \mathrm{Bord}_1 \\ {\scriptstyle \mathrm{triv}}\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{triv}} \\ \mathbb{B}G & \longrightarrow & \mathrm{Bord}_1(\mathbb{B}^{\nabla} G) \end{array} \tag{33}$$

sending $\mathbb{B}G$ to the trivial G-bundle over pt_+ together with its G-action by automorphisms. Hence, equipping a 1-dimensional topological field theory with G-symmetry in particular equips its value on a point with a G-action. Therefore a field theory with (anomalous) G-symmetry determines a finite-dimensional (projective) representation. In fact (essentially the cobordism hypothesis) this restriction determines the entire field theory.

To see this more explicitly, a connected 1-dimensional bordism with boundary is just an interval. Fixing trivializations at boundary endpoints, parallel transport identifies a G-bundle with connection over the interval with an element $g \in G$; see Figure 9. Said differently, we need the data of an ordinary topological field theory (the value of the theory on the trivial G-bundle over the interval) together with an automorphism at the boundary endpoint coming from the data already specified above by the value on the point.

It is also instructive to see what this theory assigns to closed bordisms: we obtain a function (or more generally section of a line bundle) over the moduli stack of G-bundles on the circle,

$$\mathrm{Bun}_G^{\nabla}(S^1) \simeq G/G$$

which we identify with the adjoint quotient of G acting on itself. Hence, nonanomalous theories have class functions as their partition functions, while anomalous theories have twisted class functions, by which we just mean a section of a G-equivariant line bundle on G.

Adding reflection positivity to this example forces the representations to be unitary. One way to see this geometrically is to observe that orientation reversal changes the direction of parallel transport, sending an interval whose G-bundle with connection is classified by $g \in G$ to one classified by $g^{-1} \in G$. Reflection positivity then requires the adjoint of g to be g^{-1}. This is the unitarity property.

4.2. **Twisted equivariant K-theory from 1d supersymmetric theories.** Next we investigate enhancements of Clifford anomalous $1|1$-dimensional theories (25) to include background anomalous G-symmetry. To set this up, fix a G-representation $\rho\colon G \to O(V)$; this representation will be distinct from the (complex) G-representation on $Z(\mathrm{pt}_+)$. Indeed, such a real representation gives a G-action on the Clifford algebra, which we repackage as the functor $\mathrm{Cl}(\rho)$,

$$\mathrm{Bord}_{1|1}(\mathbb{B}^{\nabla} G) \underset{\mathrm{Cl}(\rho)}{\overset{\mathbb{1}}{\overset{\curvearrowright}{\underset{\curvearrowright}{Z \Downarrow}}}} \mathrm{sBimod}\ . \tag{34}$$

$\mathrm{Bord}_1(\mathbb{B}^\nabla G)$	$\longrightarrow$	Vect
$\mathrm{pt}_+ \circlearrowleft G$	$\longmapsto$	$\mathcal{H} \circlearrowleft G,\ \rho\colon G \to \mathrm{U}(\mathcal{H})$
	$\longmapsto$	$\langle -, \rho(g)- \rangle = \langle \rho(g)^{-1}-, - \rangle \colon \overline{\mathcal{H}} \otimes \mathcal{H} \to \mathbb{C}$
	$\longmapsto$	$\rho(g)\colon \mathcal{H} \to \mathcal{H}$
	$\longmapsto$	$\mathrm{Tr}(\rho(g)) \in C^\infty(G/G)$

FIGURE 9. Some of the values of a 1-dimensional reflection positive topological theory with G-symmetry. The object pt_+ refers to the trivial G-bundle over the positively oriented point; the G-action on this object leads to a G-representation $Z(\mathrm{pt}_+) = \mathcal{H}$. Reflection positivity demands that this representation be unitary. The g labelings on the 1-dimensional cobordisms refer to the element of G gotten via parallel transport.

In other words, a G-representation ρ promotes the Clifford anomaly to an anomaly with G-symmetry. We can be very explicit about this G-action: pulling back the Pin^c-extension along the representation ρ gives a (graded) G-extension,

$$\begin{array}{ccccccccc} 1 & \to & \mathrm{U}(1) & \to & \widehat{G} & \to & G & \to & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \to & \mathrm{U}(1) & \to & \mathrm{Pin}^c(V) & \to & O(V) & \to & 1 \end{array} \tag{35}$$

A $\mathrm{Cl}(V)$-module determines a representation via the action $\mathrm{Pin}^c(V) \subset \mathrm{Cl}(V)^\times$, and so the homomorphism $\hat{G} \to \mathrm{Pin}^c(V)$ gives a projective G-action on this module.

This projective G-action appears in the value of the anomalous theory on the point: $Z(\mathrm{pt}_+) = \mathcal{H}_{\mathrm{Cl}(V)}$ is a (graded) representation of $\widehat{G}$, the pullback of the Pin^c-extension. We recognize this data as a twist for equivariant KR-theory [Fre12]. This can also be seen more abstractly following the same argument as diagram (32), and running this same argument as in the topological case the remainder of a G-symmetric anomalous theory is just a property: the operator Q generating the super semigroup representation is required to graded commute with the (projective) representation,

$$\rho(g) Q \rho(g)^{-1} = (-1)^{|g|} Q.$$

Letting (t, θ, g) denote a G-bundle over the super interval of length (t, θ) with connection whose parallel transport determines $g \in G$, a $1|1$-dimensional theory with anomalous G-symmetry is determined by the values

$$\begin{array}{rcl} \mathrm{pt}_+ & \longmapsto & \mathcal{H}_{\mathrm{Cl}(V)} \circlearrowleft \widehat{G} \\ \begin{array}{c} + \\ \big\vert {\scriptstyle (t,\,\theta,\,g)} \\ + \end{array} & \longmapsto & e^{-tH + \theta Q} U_g \\ P_g \to \bigcirc^{\mathrm{hol}_{S^1} = g} & \longmapsto & \mathrm{sTr}_{\mathrm{Cl}(V)}(e^{-tH+\theta Q} U_g) \in \Gamma(G/G; \mathrm{HH}_0(\mathrm{Cl}(V))^\vee) \end{array}$$

along with a trace class condition described in the last line. By general categorical considerations, the partition function is valued in the Hochschild homology $\mathrm{HH}_0(\mathrm{Cl}(\rho))^\vee$. We view this as a line bundle over $[G/G]$, again identified with the moduli of G-bundles with connection on the circle. For each G-bundle with connection on a super interval of length (t, θ) with $t > 0$, we get a *Clifford super trace* as indicated above, e.g., see [ST04, Definition 3.2.16]. This is as one expects from representation-theoretic considerations: characters of projective representations are *twisted* class functions, which one can repackage as sections of a line bundle over the adjoint quotient.

Problem 12 ($\star$)**.** The collection of Cl_n-anomalous theories with background G-symmetry gives geometric cocycles for G-equivariant twisted K-theory.

Remark 4.5. For certain choices of definition, the above can be made a theorem: in [HST10, Corollary 6.34], Hohnhold–Stolz–Teichner show field theories defined there assemble into a stack that can be identified with twisted equivariant K-theory in [BEG24]. But it remains an open problem to cement the link with the geometry of background gauge fields as sketched above, and ultimately this geometry needs to be better understood for the 2-dimensional enhancements described below.

As a special case, we can consider enhancing the V-family of field theories representing the Thom class (26) to a theory with anomalous G-symmetry. In this case we have

$$\mathcal{H} = \mathrm{Cl}(V)_{\mathrm{Cl}(V)} \circlearrowleft \widehat{G}, \quad Q_v \tag{36}$$

where now it is important to recognize these data as a G-equivariant family on V, or equivalently, a family over the quotient stack $[V/G]$.

Problem 13 ($\star$)**.** For $G = O(V)$ acting on V, the $V/O(V)$-family of field theories (36) gives a cocycle representative of the universal twisted Thom classes

$$\mathrm{Th}_V \in \mathrm{KR}^{\tau_V}_{O(V)}(V, V \setminus 0) \tag{37}$$

with twist $\tau_V = (\dim(V), w_1(V), W_3(V))$ determined by the dimension, 1st Stiefel–Whitney class, and 3rd integral C_2-equivariant Stiefel–Whitney class.

Remark 4.6. Again, depending on what definitions one is willing to accept, the above can be made into a real theorem: the families of Clifford modules described above are precisely the cocycle descriptions of universal Thom classes in [BEG24]. Nevertheless, there are important details to work out when writing these as functors out of a cobordism category.

The classes (37) contain all the orientations of K-theory you might want, e.g., the MU-orientation of KU, the $M\,\mathrm{Spin}^c$-orientation of KU and the $M\,\mathrm{Spin}$-orientation of KO, as well as the various (twisted) equivariant orientations of these theories.

4.3. **A roadmap for a geometric model of twisted equivariant TMF.** The 2-dimensional generalizations of Problems 12 and 13 are expected to construct twisted G-equivariant TMF and its (twisted, equivariant) string orientation. There is still a lot of work to be done, but we will sketch the picture that is emerging. We also list some important intermediate objectives (listed as Problems) that provide one possible path towards a proof of Conjecture A.

The basic objects we seek to construct are Fer_V-anomalous 2|1-dimensional field theories (28) with background (anomalous) G-symmetry,

$$\mathrm{EBord}_{2|1}(\mathbb{B}^\nabla G) \xrightarrow[\mathrm{Fer}(\rho)]{\overset{\mathbb{1}}{\overset{Z \Downarrow}{\longrightarrow}}} \mathrm{sBimod} \tag{38}$$

where Fer_ρ is the G-symmetry enhancement of Fer_V depending on a G-representation $\rho\colon G \to O(V)$. Just as in the 1-dimensional case, to start we need a G-symmetric extension of Fer_V.

This anomaly in (38) is expected to be a straightforward generalization of the chiral theory described in §3.4, where the trivial bundle V is replaced by a vector bundle associated to a G-bundle and representation. Specifically, we have the expected values

$\mathrm{Bord}_2^{\mathrm{Spin}}(\mathbb{B}^\nabla G)$	$\xrightarrow{\mathrm{Fer}_\rho}$	sBimod
$\left\{\begin{array}{c}\text{spinor bundle } \mathbb{S} \to S^1 \\ \text{+principal bundle } P \to S^1\end{array}\right\}$	$\longmapsto$	$\mathrm{Cl}(\Gamma(S^1, \mathbb{S} \otimes P \times_\rho V))$
$\left\{\begin{array}{c}\Sigma \text{ spin manifold with Dirac operator } \not{D} \\ \text{+principal bundle } P \to \Sigma\end{array}\right\}$	$\longmapsto$	$\begin{array}{ll}\mathrm{Fock}(\not{D}_{P\times_\rho V}) & \Sigma \text{ connected, } \partial\Sigma \neq \emptyset \\ \mathrm{Pf}(\not{D}_{P\times_\rho V}) & \Sigma \text{ closed.}\end{array}$

In words, the G-bundle on the circle is sent to the Clifford algebra of the associated bundle for the representation ρ. A connected cobordism with nonempty boundary is sent to the irreducible Clifford module for the Lagrangian correspondence associated with the Dirac operator twisted by the associated bundle $P \times_\rho V$. A closed cobordism is sent to the Pfaffian line for this twisted

Dirac operator. This is again a chiral theory, so it is expected to have an essentially canonical supersymmetric enhancement. This has not yet been worked out in detail.

Extending the theory down to points (either its chiral version or the supersymmetric enhancement) is already open when $G = \{e\}$. So it might seem that adding the extra G-structures in equation (38) is even harder. Instead it seems that this additional structure constrains a lot of the potential choices, making this a feature rather than a bug. Indeed, it seems that the equivariant theory is a boundary theory for a 3d theory with $N = 1$ supersymmetry, giving *even more* structure. To briefly recall this, the (nonsupersymmetric) chiral free fermion can be identified with a restriction of a three-dimensional Dai–Freed theory, whose value on a closed 3-manifold is an exponentiated η- or ξ-invariant [DF94, Jen05]. This is in parallel to (21) and Problem 9, where the Clifford algebras define a 2-dimensional theory whose restriction gives the relevant anomaly for 1|1-dimensional theories.

Problem 14 ($\star\star$)**.** Construct a 3d $N = 1$ supersymmetric enhancement of Dai–Freed theory, starting with a supersymmetric (spin) Chern–Simons theory enhancing [Jen05]. Define the anomaly theories Fer_ρ, as 2d $N = (0, 1)$ boundaries of the 3d $N = 1$ theory,

$$\mathrm{Fer}_\rho \colon \mathrm{EBord}_{2|1}(\mathbb{B}^\nabla G) \to \mathrm{sBimod}\,. \tag{39}$$

The above problem would fix a class of (invertible) anomaly theories. Next one requires a target 3-category that accommodates these anomalies *and* allows for interesting examples of anomalous field theories. At the moment, there aren't so many 3-categories with rich enough functional analysis to provide a habitable ecosystem for these examples. Bicommutant categories provide one promising new direction [Hen17].

Problem 15 ($\star\star\star$)**.** Define fully-extended Fer_ρ-anomalous theories, and unpack the geometric and analytic content of such a theory.

Just as a module for a Clifford algebra with G-action gives a projective G-representation, an (implementable) module for the infinite-dimensional Clifford algebra $\mathrm{Cl}(\Gamma(S^1, \mathbb{S} \otimes P \times_\rho V))$ gives a projective representation of the twisted loop group

$$L_P G = \{G\text{-bundle isomorphisms } P \xrightarrow{\sim} P\}$$

that when P is the trivial bundle recovers a representation of a central extension of the standard loop group $LG = C^\infty(S^1, G)$ [PS86]. After passing to the $\mathrm{Pin}(V)$-extension $\widehat{G}$ of G, we obtain a representation of $L\widehat{G}$ with level $\widehat{\rho}^* \frac{p_1}{2} \in H^4(B\widehat{G}; \mathbb{Z})$ [FHT11].

When extending down to points, one expects to enhance this loop group representation to a category with an action by the *string group* or its closely related variants coming from a (graded) categorical extension of $O(V)$ [SP11].

A final step to set up a weak version of Conjecture A is the following.

Problem 16 ($\star\star\star$)**.** Assemble the collection of fully-extended Fer_ρ-anomalous theories into a space.

Remark 4.7. Following Remark 3.20, it is likely one actually wants a stack of theories rather than an actual space, where the concordance space of the stack recovers the desired spaces in the above problem.

We rephrase Conjecture A in the language and notation from above.

Conjecture E. The collection of Fer_ρ-anomalous 2|1-dimensional field theories with G-symmetry gives geometric cocycles for G-equivariant twisted TMF.

But of course we want more than the spaces representing TMF—we want the spectrum structure. Again it seems useful to ask about the objects that have the maximal amount of structure, namely the twisted Thom classes.

Problem 17 ($\star\star\star$)**.** Construct universal twisted Thom classes.

Conjecture F. A fully extended version of Fer_ρ comes with a Thom class in twisted equivariant TMF

$$\mathrm{Th}(\rho) \in \mathrm{TMF}_G^{\tau_\rho}(V, V \setminus 0)\,.$$

This should in particular determine the string orientation.

Lurie's 2-equivariant elliptic cohomology extends equivariant elliptic cohomology of spaces to suitable higher stacks. In particular, a level $\ell \in H^4(BG;\mathbb{Z})$ determines a theta line $\mathcal{L}_\ell$ over the moduli of G-bundles on elliptic curves. This has a universal property and can be used to produce a map connecting field theories with TMF [Lur09b, §5.5].

Problem 18 ($\star\star\star$)**.** Construct 2-equivariant elliptic cohomology.

This is a problem in homotopy theory rather than one in the Stolz–Teichner program. Significant progress is underway [GM23].

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